755 Phoenix Drive, Ann Arbor MI, 48108, USA
Phone: 734 668-2930 • Fax: 734 668-2877 • Email: info@carsim.com
carsim.com

Powertrain System (Motorcycle)

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A conventional powertrain model in BikeSim includes an engine, primary gear, clutch (or torque converter), and a transmission (Figure 1).

There are two different ways to transmit the torque from the transmission to the rear wheel: one is by shaft, and another is by chain. With the shaft drive, the transmission output is applied to the end of the drive shaft, and drive torque at the wheel is calculated through the torsional stiffness, damping, and final gear (Figure 2). With the chain drive, the transmission output is applied to the gearbox sprocket, and the drive torque at the wheel is calculated by the chain mechanisms (Figure 3).

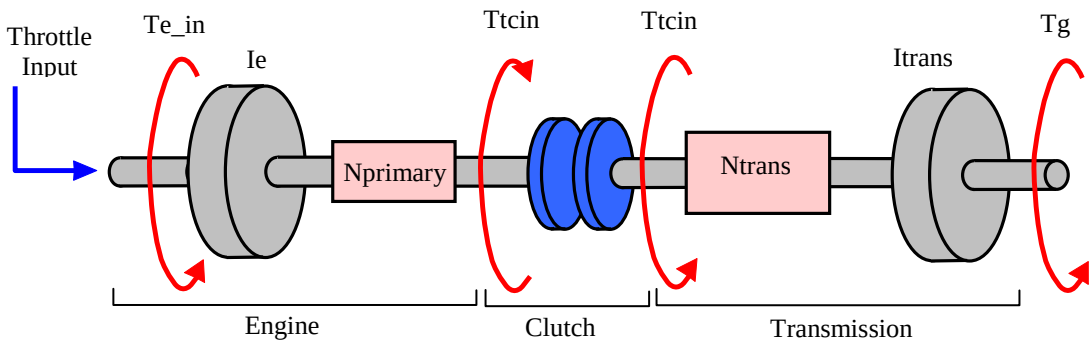


Figure 1. Torques between engine, clutch, and transmission.

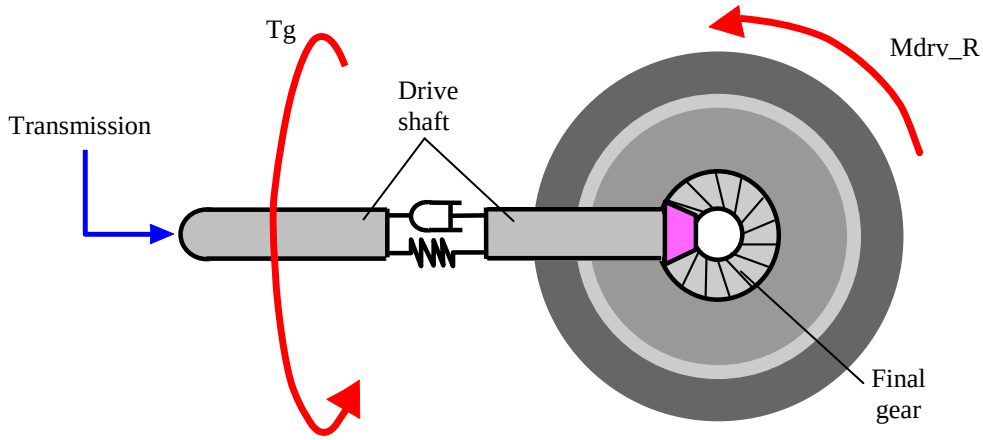


Figure 2. Shaft drive system (rear wheel).

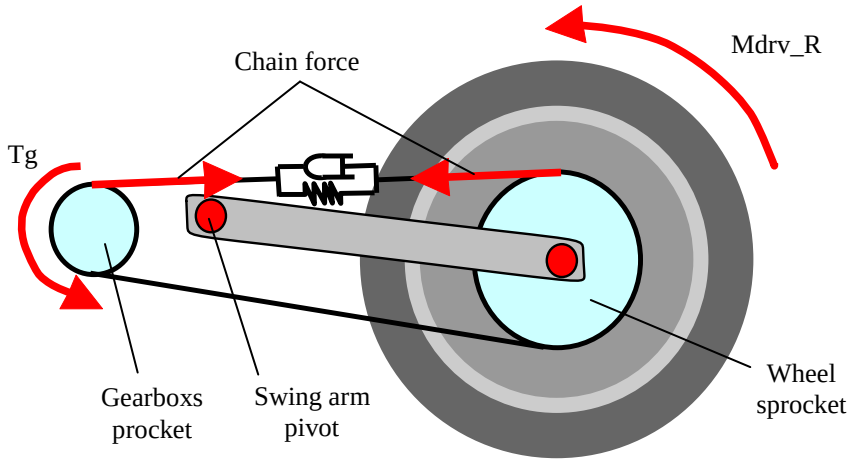


Figure 3. Chain drive system (rear wheel).

This document is the main source of information for how the conventional powertrain model is set up and used. However, two other Help documents cover other aspects of the powertrain model and its use:

- *Rider Controls* describes how inputs to the powertrain may be provided as open-loop functions of time, or using target speed, or acceleration with the closed-loop speed controller (SC) included in BikeSim. This document also covers the closed-loop clutch and throttle modulation used for shifting with a clutch automatically.
- *Powertrain for Electric and Hybrid Electric Vehicles (BEV/HEV)* describes the powertrain options for electric and hybrid vehicles.

Assembling Powertrain Components

Figure 4 shows the screen that assembles the information needed to define a shaft drive powertrain – power source (1), primary gear (or CVT) (6), clutch (or torque converter) (10), transmission (12), and drive shaft (14). Each part can be switched between the internal model and an external model by using a drop-down list (e.g., the drop-down list of internal-combustion engine model options, Figure 5).

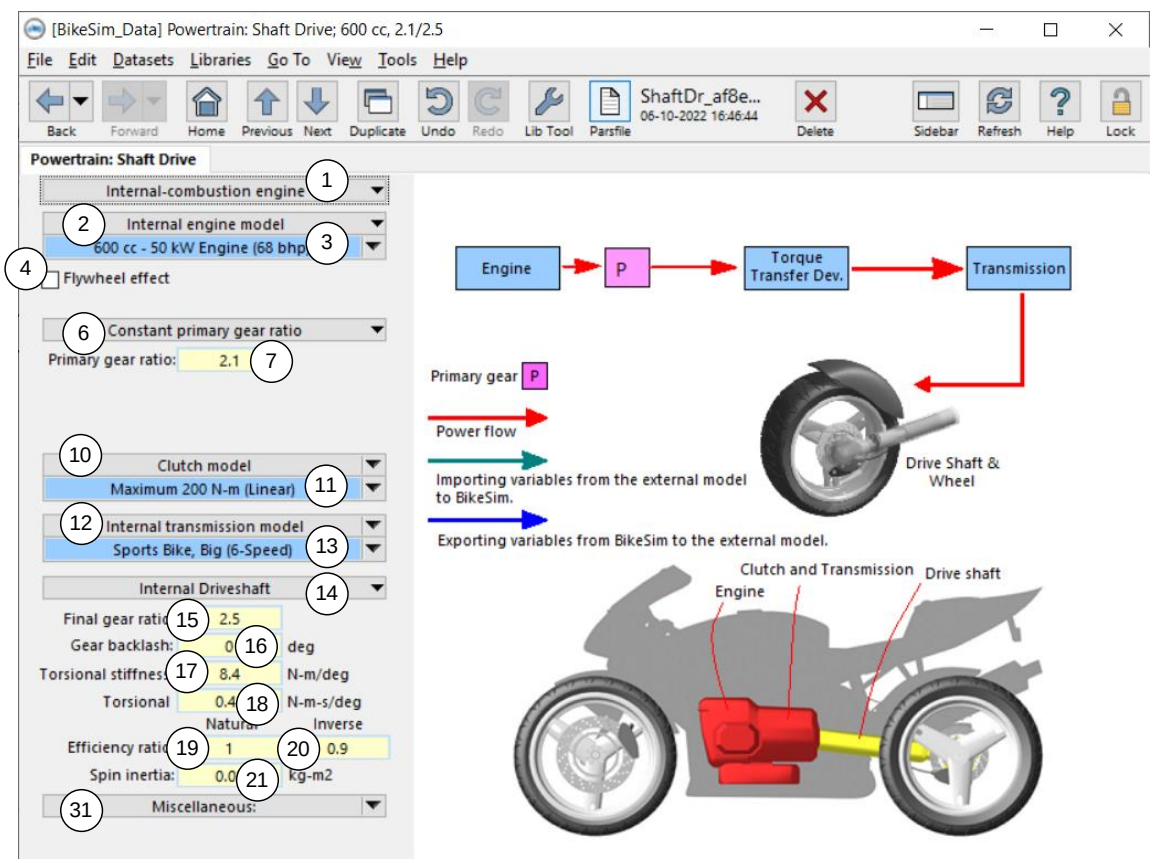


Figure 4. Powertrain assemble screen for shaft drive system.

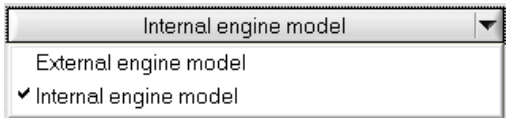


Figure 5. Drop-down list for choosing powertrain option.

The block diagram shown on the screen describes the power flow from the engine to the wheels using red arrows. If an external model is selected, the internal model is completely ignored; instead, the power flow is completely routed through the external model and the internal model does not pass any torque and physical state information (such as rotational angle and speed) to the next parts (Figure 6).

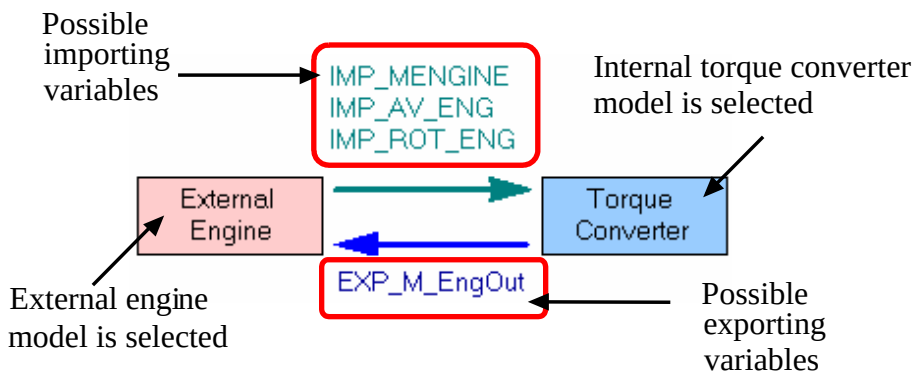


Figure 6. Diagram between the external and internal model blocks.

When the external model is selected for part of the system, you must feed power and physical state information from/to the next parts using the external import/export variables. In the block diagram on the screen, an external model is shown by a red block, and possible importing variables from the external model are shown by green and possible exporting variables to the external model are shown by blue. Those import/export variables are not set automatically; you need to set them using the **I/O Channels: Import** and **I/O Channels: Export** screens.

BikeSim support three different propulsion systems. Select the type of system from the ring control (1) on the powertrain assemble screen (Figure 7):

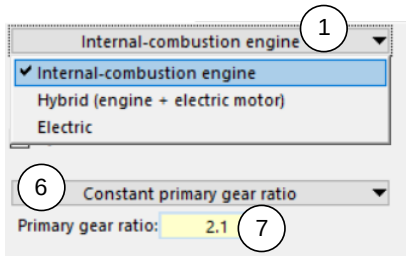


Figure 7. Select either internal-combustion, hybrid-electric or electric system.

1. **Internal-combustion engine**: an engine, a mechanical clutch and a transmission are connected in series,

2. **Hybrid (engine + electric motor)**: power source is an engine and/or a motor, they are parallelly connected or disconnected by clutch, or
3. **Electric**: the power source is only an electric motor which outputs to chain sprocket or drive shaft.

Selecting the menu item **Internal-combustion engine** (`OPT_HEV = 0`) installs conventional powertrain components such as the clutch (or torque converter) and transmission, without electric or hybrid components such as electric motor and generator.

Selecting either the menu item **Hybrid (engine + electric motor)** or **Electric** installs the hybrid/electric components with the power management control (`OPT_HEV = 4` for parallel hybrid or `2` for electric) and two library links to the clutch (11) and transmission (13) screens are replaced with the links for **Powertrain: Hybrid/Electric System** and **Hybrid/Electric Power Management control** libraries, respectively (external links (10) and (12) are hidden.)

If you select either Hybrid or Electric system, please see the separate document *Powertrain for Electric and Hybrid Electric Vehicles (BEV/HEV)* for further data settings.

The powertrain options are assembled on the screen shown in Figure 4, with the controls described in the following subsections.

Engine to transmission

(1) Drop-down list of three source of power for the system (Figure 8).

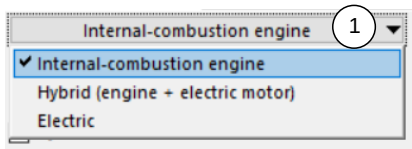


Figure 8. Three sources of power for the powertrain system.

This document covers the first option: internal-combustion engine. Information about the hybrid and fully electric options are described in the companion Help document, *Powertrain for Electric and Hybrid Electric Vehicles (BEV/HEV)*.

(2) Drop-down list for selecting an external or internal engine model (keyword = `OPT_ENGINE_INTERNAL`). The data link for the engine (3) is hidden when the external engine model is selected.

(3) Link to a **Powertrain: Engine** dataset.

(4) **Flywheel effect** checkbox (keyword = `OPT_FLYWHEEL_EFFECT`). Check to include the gyroscopic effect and inertial reaction of the engine crankshaft and flywheel. The effect is based on the engine crankshaft inertia which is set on the **Powertrain: Engine** library linked from (3). Checking the box reveals two yellow fields to define the orientation angles (yaw and pitch) of crankshaft axis (5) (Figure 9).

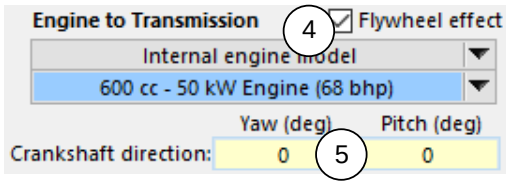


Figure 9. Crankshaft direction settings.

(5) Yaw and pitch angles of the engine crankshaft axis relative to the sprung mass coordinate system, X axis. The value of yaw angle (keyword = `A_YAW_CRANKSHAFT`) defines various types of engine layout, for example:

0 Longitudinal engine with clockwise rotating. The orientation of the crankshaft is longitudinal as perpendicular to the wheel spin axes. The crankshaft rotates clockwise looking from back.

90 Transverse engine with natural rotating. The orientation of the crankshaft is parallel to wheel spin axes. The crankshaft rotates in the same direction as the wheels.

180 Longitudinal engine with anti-clockwise rotating. The orientation of the crankshaft is longitudinal but the crankshaft rotates anti-clockwise looking from back.

270 (or -90) Transverse engine with reverse-rotating. The orientation of the crankshaft is parallel to wheels' spinning axes. However, the crankshaft rotates in the reverse direction to the wheels so that the flywheel rotation cancels out the gyroscopic effect of wheels.

The value of pitch angle (keyword = `A_PITCH_CRANKSHAFT`) defines an Euler pitch angle of the engine crankshaft axis relative to the sprung mass coordinate system. The rotational order is yaw -> pitch.

(6) Drop-down list for selecting the type of primary gear (keyword = `OPT_PRIMARY_GEAR`) (Figure 10).

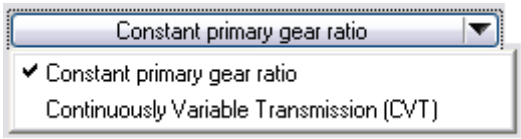


Figure 10. Drop-down options for type of primary gear.

1. When **Constant primary gear ratio** is chosen, the primary gear between the engine and clutch has a constant gear ratio whose parameter field (7) is displayed. In this case, the controls on the screen are shown to match this choice (Figure 4).
2. When **Continuously Variable Transmission (CVT)** is chosen, a data link for CVT (7) and associated parameter fields ((8) and (9)) are displayed (Figure 11) in which you can specify the CVT powertrain. In this case, the type selector and link for the ordinal transmission ((12) and (13)) are hidden and its gear ratio and efficiencies are treated as 1.0 (power just passes through the ordinal transmission location.)

(7) Primary gear information associated with type selected with the above list (6).

1. If the primary gear is **Constant primary gear ratio** as shown in Figure 4, a data field is shown with the primary gear ratio (keyword = `R_GEAR_PRIMARY`). If this gear ratio is greater than 1, then the clutch speed is smaller than the engine speed. (A typical setting is that this ratio is greater than 1.)

2. If the primary gear is **Continuously Variable Transmission (CVT)**, a link is shown to a **Powertrain: CVT Gear Ratio** dataset and several data fields are shown (⑧,⑨, Figure 11).

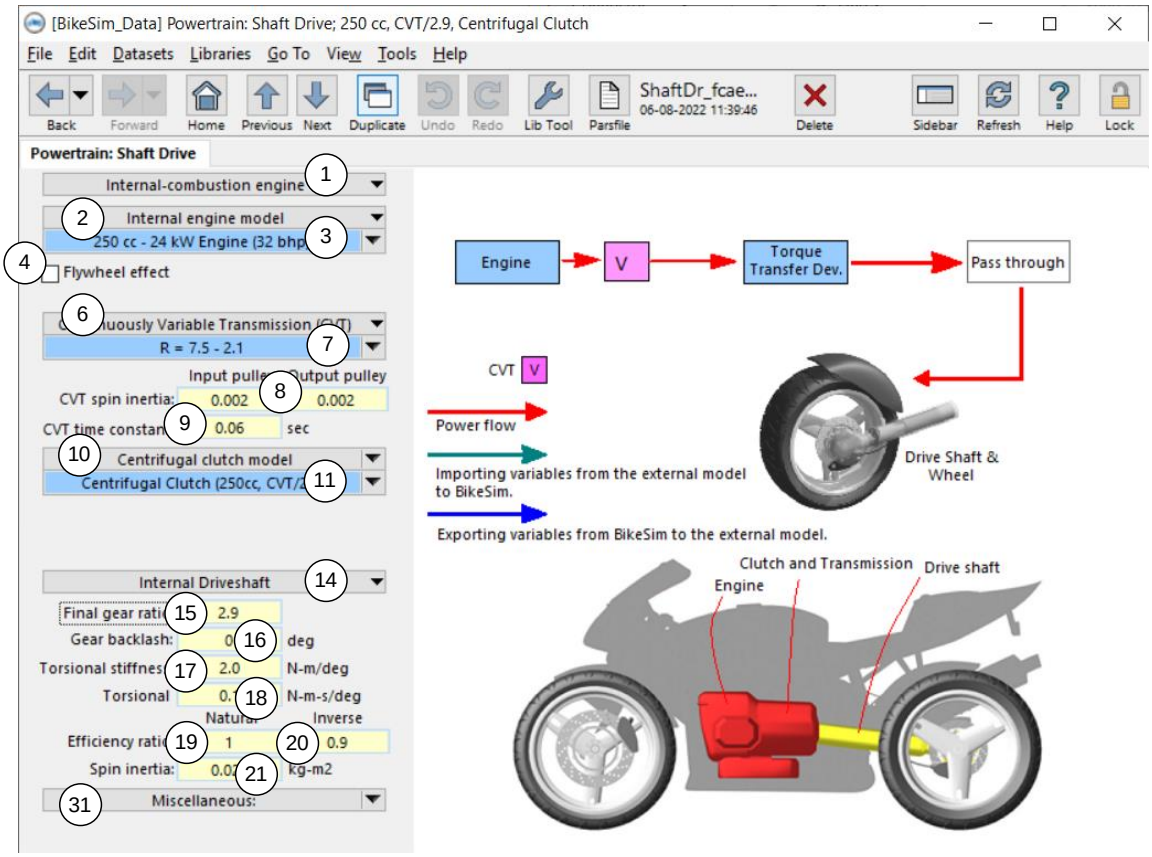


Figure 11. Powertrain assemble screen for shaft drive system and CVT.

⑧ CVT spin inertia for the input and output pulleys (keywords = `ICVT_INPUT_PULLEY` and `ICVT_OUTPUT_PULLEY`).

⑨ Dynamic time constant for CVT gear ratio (keyword = `TC_RGEAR_CVT`).

⑩ Drop-down list for selecting the type of torque coupling device (Figure 12, keyword = `OPT_PWR_CPL_INTERNAL`). The data link for the torque coupling ⑪ is hidden when the external torque coupling model is selected.

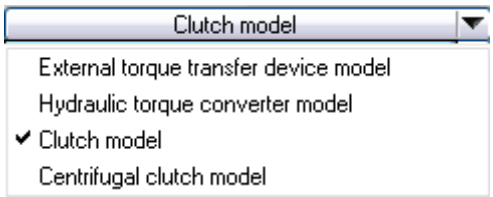


Figure 12. Drop-down list for selecting the torque coupling option.

(11) Link to either a **Powertrain: Clutch Torque** dataset, a **Powertrain: Torque Converter** dataset or a **Powertrain: Centrifugal Clutch Torque** associated with type selected with the above list (10).

> **Note** Most CVT powertrains involve a centrifugal clutch system. Therefore, when you select CVT in the drop-down list (6), you might want to select **Centrifugal clutch model** in the drop-down list (10).

(12) Drop-down list for selecting an external or internal transmission model (keyword = `OPT_TRANS_INTERNAL`).

(13) Link to either a **Powertrain: Transmission (External)** dataset or a **Powertrain: Transmission** dataset.

Drive shaft

The parameters and summary information for the drive shaft are listed in the section of the Echo file shown in Figure 13.

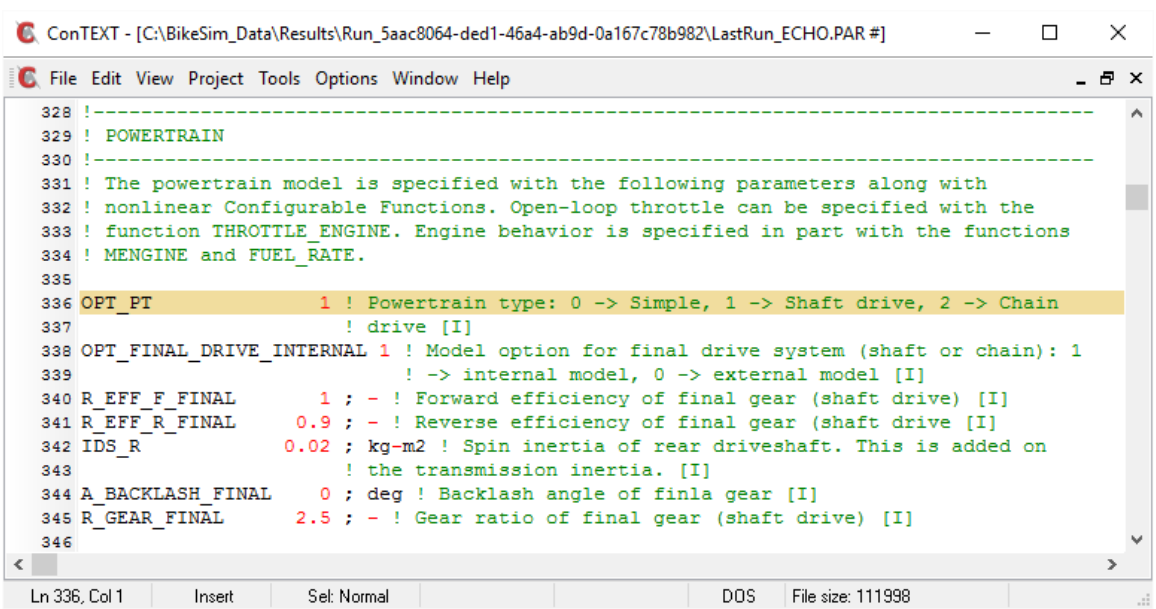


Figure 13. Drive shaft parameters in the Echo file

(14) Drop-down list for selecting an external or internal shaft drive model (keyword = `OPT_FINAL_DRIVE_INTERNAL`). The parameter fields for the drive shaft ((15) - (21)) are hidden when the external drive shaft model is selected.

(15) Final gear ratio (keyword = `R_GEAR_FINAL`). This gear is located between the transmission and wheel.

(16) Gear backlash angle (keyword = `A_BACKLASH_FINAL`). The gear backlash expresses the mechanical delay during the transition between acceleration and engine braking.

(17) Torsional spring rate of the drive shaft (keyword = `M_SHAFT_ROT_COEFFICIENT`).

(18) Torsional damping coefficient of the drive shaft (keyword = `M_SHAFT_AV_COEFFICIENT`).

> **Note** The linear torsional spring rate and damping coefficient can be replaced by tables that define nonlinear relationships between relative twist and torque, or twist rate and torque. Those nonlinear tables can be set with the **Generic Table** screen. Datasets for several tables can then be assembled with the **Generic Data Group** screen and linked from the miscellaneous link (31) on this screen. The keywords for the tables are `M_SHAFT_ROT_TABLE` (twist effect) and `M_SHAFT_AV_TABLE` (twist rate).

(19) Natural (forward) efficiency for driveshaft (keyword = `R_EFF_F_FINAL`). This efficiency is used when the engine drives the wheel.

(20) Inverse (reverse) efficiency for driveshaft (keyword = `R_EFF_R_FINAL`). This efficiency is used when the wheel drives the engine.

(21) Driveshaft inertia (keyword = `IDS_R`). The parameter must have a positive non-zero value.

Drive Chain

On the **Powertrain: Chain Drive** screen, the drive shaft parameters ((14) - (21)) are replaced by the drive chain parameters ((22) - (30)) as shown in Figure 14.

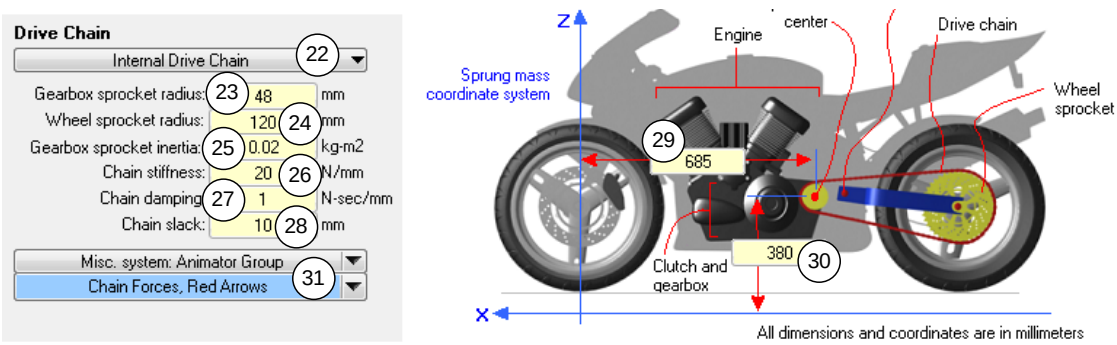


Figure 14. Screen for the chain drive mechanisms.

The parameters and summary information for the drive chain are listed in a section of the Echo file as shown in Figure 15.

```
!-------------------------------------------------------------------------------
! POWERTRAIN
!-------------------------------------------------------------------------------
! The powertrain model is specified with the following parameters along with
! nonlinear Configurable Functions. Open-loop throttle can be specified with the
! function THROTTLE_ENGINE. Engine behavior is specified in part with the functions
! MENGINE and FUEL_RATE.

OPT_PT                2 ! Powertrain type: 0 -> Simple, 1 -> Shaft drive, 2 -> Chain
                        ! drive [I]
OPT_FINAL_DRIVE_INTERNAL 1 ! Model option for final drive system (shaft or chain): 1
                           ! -> internal model, 0 -> external model [I]
K_CHAIN_STRETCH      20 ; N/mm ! Chain stiffness
D_CHAIN_STRETCH       1 ; N-sec/mm ! Chain damping

!-------------------------------------------------------------------------------
! POWERTRAIN CHAIN DRIVE SYSTEM
!-------------------------------------------------------------------------------
H_SPRKT_GR          400 ; mm ! Height (Z coord) of gearbox sprocket (only chain drive)
                        ! [I]
IYY_SPRKT_GR       0.02 ; kg-m2 ! Spin inertia of gearbox sprocket (only chain drive)
                        ! [I]
L_SLACK_CHAIN        10 ; mm ! Chain slack (only chain drive)
LX_SPRKT_GR         800 ; mm ! X distance gearbox sprocket is behind vehicle origin
                        ! (only chain drive) [I]
RAD_SPRKT_GR         48 ; mm ! Radius of gearbox sprocket (only chain drive) [I]
RAD_SPRKT_WL        120 ; mm ! Radius of rear wheel sprocket (only chain drive) [I]
```

Figure 15. Drive chain parameters in the Echo file (edited).

(22) Drop-down list for selecting an external or internal drive chain model (keyword = `OPT_FINAL_DRIVE_INTERNAL`). The parameter fields for the drive chain ((26) - (28)) are hidden when the external drive chain model is selected.

> **Note** The sprocket radiuses and inertia fields ((23) - (25)) still remain even though the external drive chain is selected in order to calculate chain kinematics.

(23) Gearbox sprocket radius (keyword = `RAD_SPRKT_GR`).

(24) Wheel sprocket radius (keyword = `RAD_SPRKT_WL`). Wheel sprocket radius divided by gearbox sprocket radius is a final gear ratio.

(25) Gearbox sprocket inertia (keyword = `IYY_SPRKT_GR`).

(26) Chain stiffness (keyword = `F_CHAIN_X_COEFFICIENT`).

(27) Chain damping (keyword = `F_CHAIN_VX_COEFFICIENT`).

> **Note** The linear stiffness and damping coefficient can be replaced by tables that define nonlinear relationships between chain stretch and tensional force, or chain stretch rate and tensional force. Those nonlinear tables can be set with the **Generic Table** screen. Datasets for several tables can then be assembled with the **Generic Data Group** screen and linked from the

miscellaneous link (31) on this screen. The keywords for the tables are `F_CHAIN_X_TABLE` (stretch effect) and `F_CHAIN_VX_TABLE` (stretch rate). However, only the positive tensional force is generated when the chain is stretched (Figure 3).

(28) Chain slack (keyword = `L_SLACK_CHAIN`). The slack expresses the mechanical delay during the transition between acceleration and engine braking.

(29) Distance that the center of gearbox sprocket lies behind the front axle (keyword = `LX_SPRKT_GR`)

(30) Height of the center of gearbox sprocket above the sprung mass origin at the design load condition with zero pitch and roll (keyword = `H_SPRKT_GR`). The locations of gearbox sprocket and wheel center determine how much the dive or anti-dive force is produced due to the tension of the chain.

Miscellaneous link

(31) Miscellaneous link. This has no predefined purpose, but can be linked to a data set that specifies parameters for extensions to the model, or information for the animation namely displaying chain forces arrows.

Engine and Primary Gear

The parameters and summary information for the engine are listed in the section of the Echo file shown in Figure 16.

```
342 !-----------------------------------------------------------------------------------
343 ! ENGINE
344 !-----------------------------------------------------------------------------------
345 INSTALL_ENGINE         ! VS Command to install the supporting features for an engine
346                        ! in the powertrain
347
348 OPT_ENGINE_INTERNAL 1 ! Engine model: 1 -> internal, 0 -> external [I]
349 OPT_ENGINE_RUNNING  1 ! [D] Is engine running? 1 -> yes, 0 -> no
350 OPT_PRIMARY_GEAR    0 ! Model option for the gear ratio after the engine: 0 ->
351                       ! constant primary gear ratio, 1 -> continuously variable
352                       ! (CVT) [I]
353 OPT_THROTTLE_DELAY  1 ! Use throttle time constants: 0 -> no, 1 -> yes [I]
354 AV_ENG_IDLE       938 ; rpm ! Engine idle speed [I]
355 ENGINE_ESC_PG       5 ; 1/s ! [D] P gain of ESC engine torque control [I]
356 ENGINE_ESC_IG     0.5 ; 1/s2 ! [D] I gain of ESC engine torque control [I]
357 ENGINE_STALL_DAMP 0.2 ; N-m-s/deg ! [D] Damping rate of the stalled engine [I]
358 IENG            0.034 ; kg-m2 ! Spin inertia of engine crankshaft [I]
359 ITC_INPUT_SHAFT     0 ; kg-m2 ! Spin inertia of input shaft of torque converter [I]
360 R_GEAR_PRIMARY    2.1 ; - ! Gear ratio of primary gear [I]
361 TC_TH_APP        0.12 ; s ! Time constant for engine throttle application
362 TC_TH_RELEASE    0.05 ; s ! Time constant for engine throttle release
363 OPT_FLYWHEEL_EFFECT 0 ! Gyroscopic effect of engine crankshaft and flywheel: 1 ->
364                       ! include , 0 -> not include [I]
```

Figure 16. Engine parameters in the Echo file.

The engine is the power source of the vehicle when the driveline model is engaged. Engines are typically characterized by the flywheel torque measured on a dynamometer as a function of throttle position, *throttle*, and rotational speed, ω_e. These properties are defined in a nonlinear table that relates torque to engine speed for different throttle settings, such as:

$$T_{e_in} = f_{et}(throttle, \omega_e) + T_{e_control} \tag{1}$$

where $T_{e_control}$ is an additional engine torque which is modulated by ESC/TCS controller and its details are described in the following subsection (page 13).

The relationship between the engine torque (T_{e_in}), engine moment of inertia (I_e), engine angular acceleration (α_e), primary gear ratio ($N_{primary}$), moment of inertia of the clutch on the engine side (I_{tc_in}), and clutch input torque (T_{tcin}) is:

$$\begin{aligned} & I_{e_comb} \times \alpha_e = T_{e_in} - T_{e_out} \\ & \text{where} \quad T_{e_out} = T_{tcin} / N_{primary}, \\ & \qquad I_{e_comb} = I_e + (I_{tc_in} / N_{primary}{}^2). \end{aligned} \tag{2}$$

Engine angular speed and rotation angle (ϕ_e) are obtained by integrating the differential equation:

$$\dot{\omega}_e = \alpha_e = \frac{T_{e_in} - T_{e_out}}{I_{e_comb}} \tag{3}$$

$$\dot{\phi}_e = \omega_e \tag{4}$$

The above parameters and table function involved in these engine calculations are set on the screen in Figure 17, using the controls described below.

① Checkbox for using a first-order time delay for the engine throttle. When checked, extra data fields (② and ③) are shown that specify the time constants. If not checked, the engine throttle responds instantly without delay.

② Dynamic time constant for opening the engine throttle (keyword = `TC_THR_APP`).

③ Dynamic time constant for closing the engine throttle (keyword = `TC_THR_REL`).

④ Optional link to a **Powertrain: Fuel Consumption Rate** dataset defining the rate of fuel consumption for this engine. When linked, the fuel consumption rate (kg/sec) and total fuel consumption (kg) for the run are calculated.

⑤ Engine rotational inertia at crankshaft (keyword = `IENG`). This parameter includes the inertia properties of all parts that spin with the engine output shaft and possibly portions of the hydraulic torque converter.

⑥ Idle speed (keyword = `AV_ENG_IDLE`). This is the static engine rotational speed when there is zero throttle input.

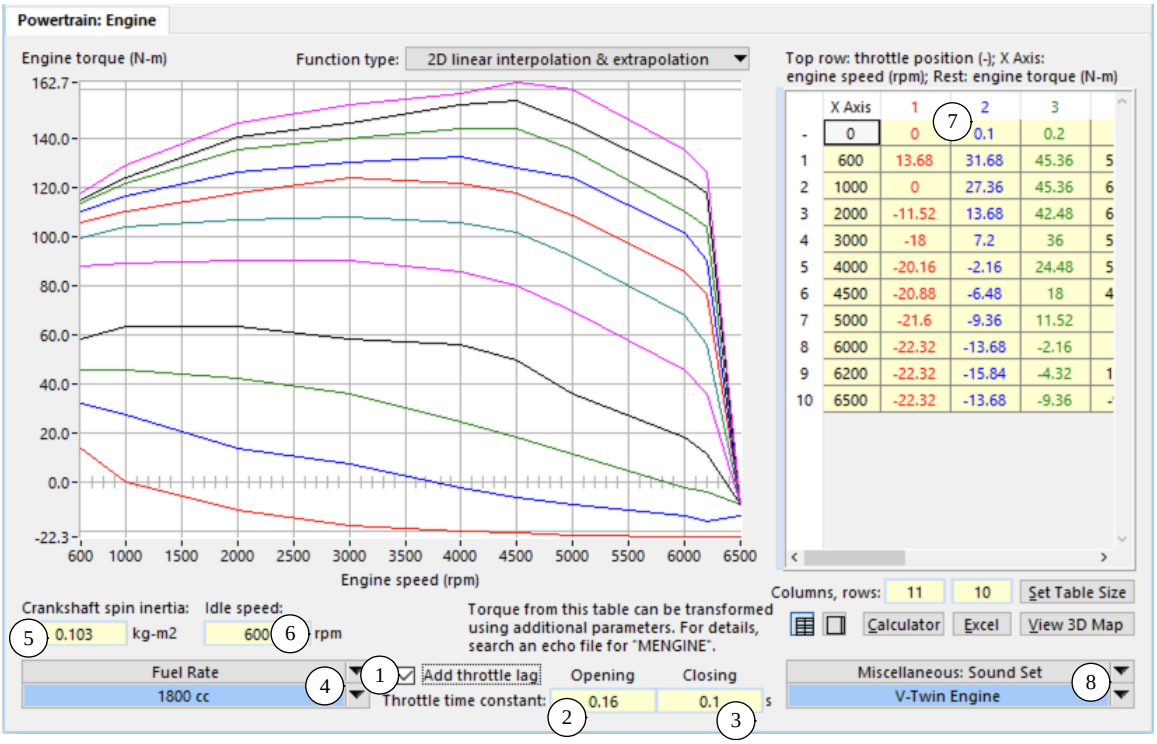


Figure 17. Screen for an engine.

⑦ Tabular data for engine torque (root keyword = `MENGINE`).

⑧ Link to **Genetic Data List** dataset which involves an engine sound profile. This is not used by the math model but is used by the animation if provided.

ESC/TCS torque control interface

ESC (Electric Stability Control) or TCS (Traction Control System) is required by regulations for all passenger cars since 2012. Some commercial ESC systems for motorcycles (also referred to as ASC/MSC) have been introduced for improving stability. In these controllers, the engine torque is often modulated to minimize an excessive engine torque to prevent an instability due to wheel slipping. Further, some autonomous throttle control systems assist when climbing gravel hills.

With these controllers in mind, VS powertrain includes a controllable engine torque $T_{e_control}$ in Equation 1 (page 12).

The ESC/TCS controller typically commands a demand torque to the engine based on the driver requesting engine torque and some vehicle operating states, such as speed, yaw rate, wheel slip ratio, etc.

The VS model does not have a built-in ESC/TCS controller. A control algorithm defined with VS Commands or from an external model such as Simulink, actual hardware ECU, etc., provides the torque request $T_{e_ESC_rq}$, which is accessed in the VS Solver through the import variable `IMP_MENGINE_ESC_REQUEST`.

The engine torque requested by the driver model ($T_{e_DRV_rq}$) is exportable from the VS Solver to outside controller through an output variable (`M_Eng_Rq`) and is calculated based on the current engine speed with throttle level, such as:

$$T_{e_DRV_rq} = f_{et}(throttle, \omega_e) \tag{5}$$

The VS model has an internal P+I controller which modulates the engine torque, $T_{e_control}$, in Equation 1 to meet the ESC's torque demand. The internal controller has three different operational states distinguished by an import variable (`IMP_ESC_ENGINE_CON_STATE`), 0: no control as default; 1: reduce; or 2: increase the engine torque to meet the torque request from ESC. For example, if the ESC is the reduce state (`IMP_ESC_ENGINE_CON_STATE = 1`), the ESC's target is the smaller of ESC and driver demand, such as:

$$\begin{aligned} &if\ (ESC_state = 1) \\ &\quad T_{e_target} = \min\{T_{e_ESC_rq}, T_{e_DRV_rq}\} \\ &else\ if\ (ESC_state = 2) \\ &\quad T_{e_target} = \max\{T_{e_ESC_rq}, T_{e_DRV_rq}\} \\ &else\ if\ (ESC_state = 0) \\ &\quad T_{e_target} = T_e \end{aligned} \tag{6}$$

The time derivative of the engine control torque is:

$$\dot{T}_{e_control} = \left(T_{e_target} - T_e\right) G_{P_ESC} + \int \left(T_{e_target} - T_e\right) dt\ G_{I_ESC} \tag{7}$$

All import and export variables related with the ESC torque control are summarized in Table 1 and Table 2, respectively. PI parameters are summarized in Table 3.

Table 1. Import variables related with ESC/TCS torque control

Keyword	**Notation**	**Description**
`IMP_ESC_ENGINE_CON_STATE`	*ESC_state*	ESC engine torque control state
`IMP_MENGINE_ESC_REQUEST`	$T_{e_ESC_rq}$	Engine torque requested by ESC/TCS

Table 2. Export variables related with ESC/TCS torque control

Keyword	**Notation**	**Description**
`ESC_Stat`	*ESC_state*	ESC engine torque control state
`M_Eng_Rq`	$T_{e_DRV_rq}$	Driver requesting engine torque
`M_EngEsc`	T_{e_target}	ESC targeting engine torque
`RgearDLT`	---	Driveline total gear ratio

Table 3. PI parameters for ESC/TCS torque control

Keyword	**Notation**	**Unit**	**Description**
`ENGINE_ESC_PG`	G_{P_ESC}	1/s	P gain of engine control
`ENGINE_ESC_IG`	G_{I_ESC}	1/s2	I gain of engine control

Note Some ESC/TCS control systems use driver or ESC torque request signal at the wheels instead of the engine. In such cases, users can use an export variable `RgearDLT`, which is the total gear ratio from the engine to rear wheel through a primary gear, transmission, and final gear (either shaft drive or chain drive gear ratio).

Mechanical Clutch (including Centrifugal Clutch)

Torque is transmitted from the engine to the transmission by either a hydraulic torque converter or a mechanical clutch. The parameters and summary information for those torque transfer device are listed in a section of the Echo file as shown in Figure 18.

In BikeSim model, the same equations are adopted for the mechanical clutch, centrifugal clutch and lockup clutch on the hydraulic torque converter.

Condition of slip or locked

The clutch has two states: slipping or locked. Between those two states, the equations to calculate the torque and degree-of-freedom are different. Firstly, those states are distinguished by the following operating conditions.

$$\begin{aligned}
&\text{if currently slipping then}\\
&\quad \text{If } \mathrm{abs}(T_{clutch_cap}) > \mathrm{abs}(T_{clutch_lock}) \quad AND \quad (d\omega_{clutch} \times d\omega_{clutch_old}) < 0.0\\
&\quad \text{then} \quad \text{'chage to locked'} \quad \text{else} \quad \text{'keep slipping'}\\
&\text{else (currently locked)}\\
&\quad \text{If } \mathrm{abs}(T_{clutch_cap} \times 1.02) > \mathrm{abs}(T_{clutch_lock})\\
&\quad \text{then} \quad \text{'keep locked'} \quad \text{else} \quad \text{'change to slipping'}
\end{aligned} \tag{8}$$

Where $d\omega_{clutch}$ is the speed differential between input and output plates of the clutch (ω_{tcin}-ω_{tco}) and $d\omega_{clutch_old}$ is $d\omega_{clutch}$ from one calculation step ago. The term of the condition "($d\omega_{clutch_old}$ x $d\omega_{clutch}$) < 0.0" represents that the speed differential of the clutch reverses to the other direction.

T_{clutch_cap} denotes the torque capacity of clutch. If the clutch is controlled by rider, the torque capacity is defined by the table function as:

$$T_{clutch_cap} = f_{clutch_cap}(Clutch_displacement) \tag{9}$$

```
!-------------------------------------------------------------------------------
! TORQUE TRANSFER DEVICE
!-------------------------------------------------------------------------------
! Transfer of power from the engine to the transmission is specified with the
! following parameters along with the functions CLUTCH_TORQUE, CLUTCH_TORQUE_SPIN,
! INV_CAP_TC, and RM_TC. Open-loop clutch control can be specified with the
! function CLUTCH_CONTROL.

INSTALL_TORQUE_TRANSFER_DEVICE  ! VS Command to install the supporting features for
                                ! a torque transfer  device (clutch or torque
                                ! converter) in the powertrain
OPT_CLUTCH          1 ! Torque transfer to transmission: 0 -> hydraulic torque
                      ! converter, 1 -> mechanical clutch, 2 -> torque converter
                      ! with lock-up clutch, 3 -> centrifugal clutch [I]
OPT_CLUTCH_MODE     1 ! Clutch control mode: 0 -> open-loop, 1 -> closed-loop (based
                      ! on shifting) [I]
OPT_CLUTCH_DELAY    0 ! Use clutch torque time constants: 0 -> no, 1 -> yes [I]
OPT_PWR_CPL_INTERNAL 1 ! Internal power coupling model (torque converter and
                       ! clutch): 1 -> internal, 0 -> external [I]
T_CL_HALF_HOLD  0.125 ; s ! Time spent holding the clutch at half pressure during
                      ! clutch release [I]
T_CL_HALF_TOTAL 0.375 ; s ! Time spent releasing the clutch during gear shift [I]
T_CL_PRESS_HOLD   0.1 ; s ! Time spent holding the clutch at full pressure during
                      ! gear shift [I]
T_CL_PRESS_TOTAL 0.125 ; s ! Time spent applying the clutch to full pressure during
                       ! gear shift [I]
T_GEAR_LAG       0.05 ; s ! Time lag in shifting [I]
T_TH_RETURN     0.375 ; s ! Time spent returning to original throttle during gear
                      ! shift [I]
T_TH_ZERO_HOLD    0.1 ; s ! Time spent holding zero throttle during gear shift [I]
T_TH_ZERO_TOTAL 0.125 ; s ! Time spent going to zero throttle and holding it during
                      ! gear shift [I]
VLOW_CLUTCH         5 ; km/h ! Press the clutch pedal when average of wheel speeds
                      ! is below this speed while deccelerating with brake during
                      ! the simulation, or accelerating from stop when the
                      ! simulation starts. [I]
```

Figure 18. Parameters for the torque transfer device in the Echo file.

where *Clutch_displacement* is the input value (0 – 1) from the rider control. In the case of centrifugal clutch, the clutch capacity is defined by a function of the clutch input speed as:

$$T_{clutch_cap} = f_{clutch_cap_spin}(\omega_{tcin}) \tag{10}$$

T_{clutch_lock} denotes the torque acting at the locked clutch, whose value is calculated by:

$$T_{clutch_lock} = \frac{T_{e_in} \times N_{primary} - \frac{T_g}{N_{trans}} \cdot \frac{I_{cl_in}}{I_{cl_out}}}{1 + \frac{I_{cl_in}}{I_{cl_out}}} \tag{11}$$

where I_{cl_in} denotes the total inertial moment from engine to the clutch input plate expressed at the clutch position, while I_{cl_out} denotes the total inertial moment from the clutch output plate to transmission gear box sprocket (for chain drive) or to half of drive shaft (for driveshaft) expressed at the clutch position. They are expressed as:

$$
\begin{aligned}
I_{cl_in} &= I_e \times N_{primary}{}^2 + I_{tc_in}, \\
I_{cl_out} &= \frac{I_{trans} + I_{trans_drv}}{N_{trans}{}^2} + I_{tc_out}
\end{aligned}
\tag{12}
$$

where I_{trans_drv} is the inertia of a part of drive system which is either the gearbox sprocket (for chain drive) or a half of the driveshaft (for shaft drive). See more details in the sections **Shaft Drive Model** and **Chain Drive Model**.

> **Note** The derivation of the torque load on the locked clutch is described in the Appendix (page 35).

The reason why the clutch torque capacity is multiplied by 1.02 in the currently locked condition of Equation 8 is that the static friction is considered 2% greater than the dynamic friction.

Overall, when the clutch is slipping, if the torque capacity (T_{clutch_cap}) is bigger than the torque load (T_{clutch_lock}) and the speed differential between the input/output clutch plates is reversed to the other direction then the clutch is locked. While the clutch is locked, if the torque load (T_{clutch_lock}) exceeds more than 2% of the torque capacity (T_{clutch_cap}), the clutch starts to slip.

Torque calculation and reduction of degree-of-freedom

When the clutch is in slip

If the clutch is slipping, the torque reacting to the engine (T_{tcin}) and the torque to the transmission input shaft (T_{tcout}) are the same, such as:

$$
T_{tcin} = T_{tcout} = \mathrm{sign}\left(T_{clutch_cap}, d\omega_{clutch}\right) \tag{13}
$$

where `sign` is the sign function that the sign of the first term (T_{clutch_cap}) depends on the second term ($d\omega_{clutch}$). When $d\omega_{clutch}$ is positive, T_{tcin} is also positive in this case.

When the clutch is locked (*removed degree-of-freedom of engine*)

If the clutch is locked, the clutch torque is treated as same as engine torque, the degree-of-freedom of the engine crankshaft is removed, and both sides of the clutch are treated as “one unit”. Therefore, Equation 3 and 4 are ignored and angular acceleration and angular speed of engine are treated as:

$$
\begin{aligned}
T_{tcin} &= T_{tcout} = T_e \times N_{primary}, \\
\dot{\omega}_e &= \alpha_e = \dot{\omega}_{tco} \times N_{primary}, \\
\omega_e &= \omega_{tco} \times N_{primary}.
\end{aligned}
\tag{14}
$$

However, the clutch torque for plot is compensated by the engine inertial term only when the clutch is locked such as:

$$T_{tcin}{}^{*} = T_{tcout}{}^{*} = \left\{T_e - (I_e + I_{tc_in} / N_{primary}{}^{2})\alpha_e\right\} \cdot N_{primary}. \tag{15}$$

> **Note** $T_{tcout}{}^{*}$ is also used for the criteria of driving or coasting efficiencies of the transmission, transfer cases and differentials.

User Controls

Figure 19 shows the screen for defining the properties of a mechanical clutch.

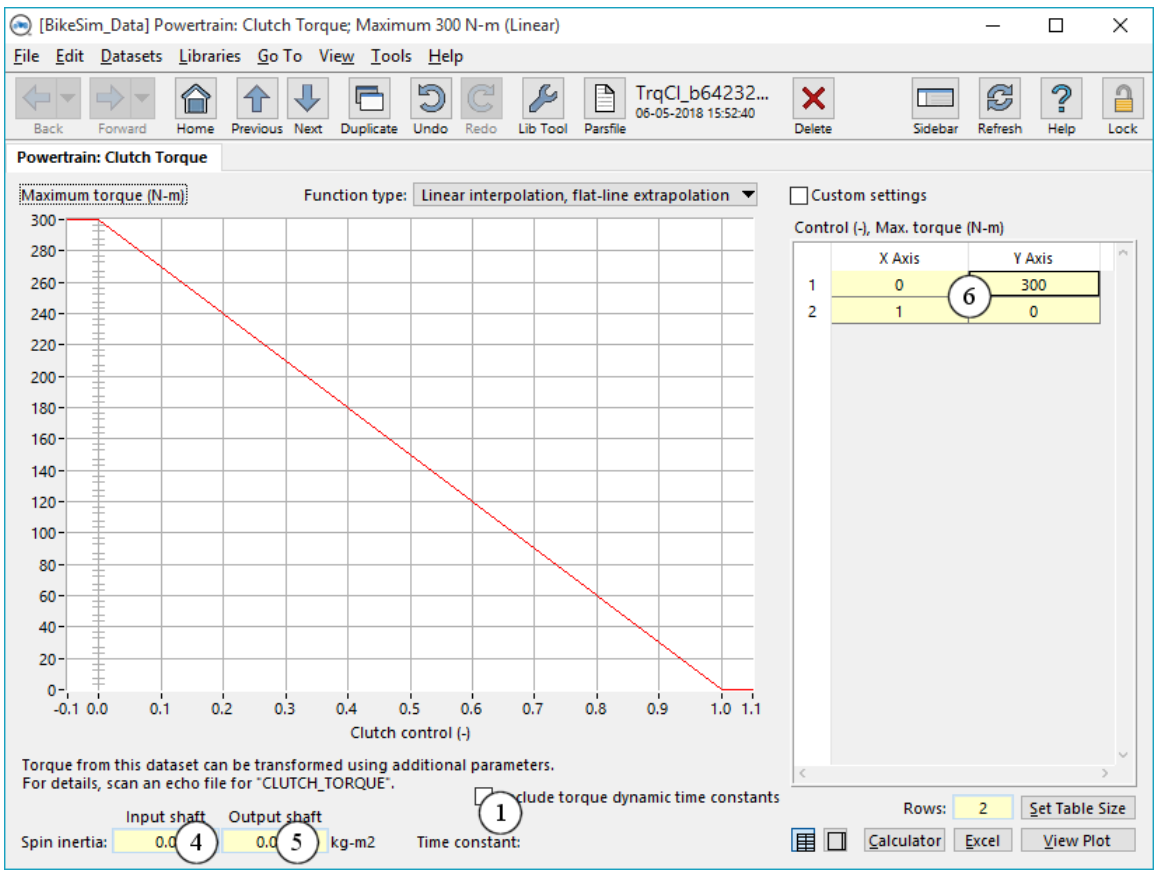


Figure 19. Screen for a mechanical clutch.

① Checkbox for using a first-order time delay of the clutch torque capacity (keyword = `OPT_CLUTCH_DELAY`). When checked, extra data fields are shown that specify the time constants (② and ③, Figure 20). If not checked, the clutch torque responds instantly without delay.

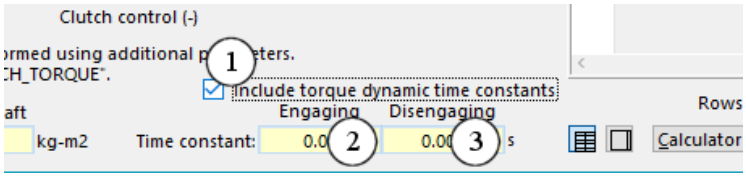


Figure 20. Two data fields are shown when the checkbox for time constants is checked.

② Dynamic time constant for engaging the clutch (keyword = `TC_CLUTCH_ENGAGE`).

③ Dynamic time constant for disengaging the clutch (keyword = `TC_CLUTCH_DISENGAGE`).

④ Spin inertia of the input shaft of the clutch (keyword = `ITC_INPUT_SHAFT`).

⑤ Spin inertia of the output shaft of the clutch (keyword = `ITC_OUTPUT_SHAFT`).

⑥ Tabular data for the maximum torque available from a mechanical clutch as a function of a rider control (root keyword = `CLUTCH_TORQUE`), or as a function of the clutch input speed (root keyword = `CLUTCH_TORQUE_SPIN.`) In the case of the rider control input, when the control is 1 (or greater), the clutch disengages such that no torque is transferred between the two shafts.

Torque Converter

Torque is transmitted from the engine to the transmission by either a hydraulic torque converter or a mechanical clutch.

If the hydraulic torque converter is selected, the torque at the output side of a torque converter (T_{tcout}) is given by the characteristic table (f_{tr}), which is the torque ratio (T_{tcout} / T_{tcin}) as a function of speed ratio (S_{tc}), multiplying the torque at the input side.

$$T_{tcout} = f_{tr}(S_{tc}) \times T_{tcin} \tag{16}$$

Torque at the input side of the torque converter (T_{tcin}) is equal to the torque load on the engine given by the characteristic table (f_{tki}), which is the inverse torque capacity (1/K) as a function of the speed ratio, (S_{tc} = ω_{tco} / ω_{tcin}), multiplying angler speed of engine.

$$T_{tcin} = sign\{f_{tki}(S_{tc})\} \times \{f_{tki}(S_{tc}) \times \omega_{tcin}\}^2 \tag{17}$$

In general, the torque converter is characterized by the torque capacity (K). However, to account for engine braking, it is difficult to describe the model directly in terms of K because normally $K \rightarrow \infty$ as $S_{tc} \rightarrow 1$. Therefore, the capacity is described by a table for $1/K$ as a function of S_{tc}. When $S_{tc} > 1$, the sign of K is assumed to be negative, thus exerting a retarding torque on the transmission, and ultimately, the wheels.

The above parameters and table function involved in these engine calculations are set on the screen in Figure 21, using the screen items described below.

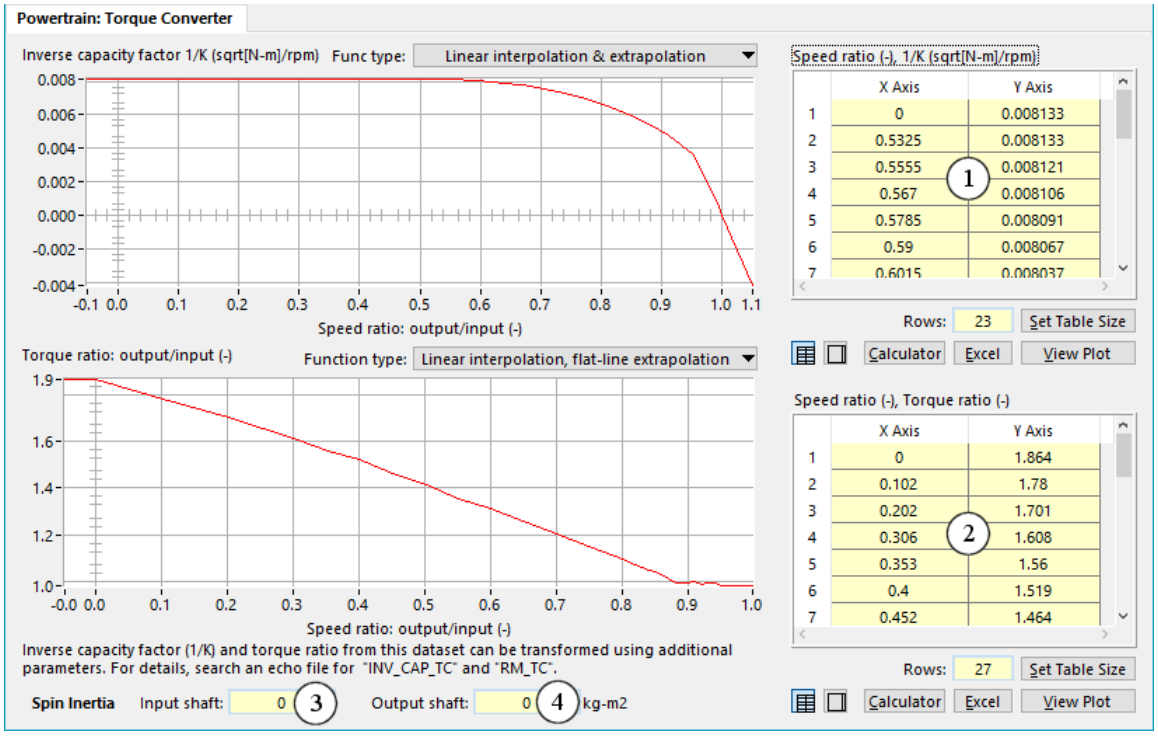


Figure 21. Screen for a torque converter.

① Inverse of the torque converter capacity as a function of speed ratio (root keyword = `INV_CAP_TC`).

② Torque amplification as a function of speed ratio (root keyword = `RM_TC`).

③ Spin inertia of the input shaft of the torque converter (keyword = `ITC_INPUT_SHAFT`).

④ Spin inertia of the output shaft of the torque converter (keyword = `ITC_OUTPUT_SHAFT`).

Transmission

The parameters and summary information for the transmission are listed in the section of the Echo file shown in Figure 22.

```
!-------------------------------------------------------------------------------
! TRANSMISSION
!-------------------------------------------------------------------------------
! The transmission is specified with the following parameters and the functions
! DOWNSHIFT_TRANS, UPSHIFT_TRANS, LOCK_AT, UNLOCK_AT, R_GEAR_CVT, R_EFF_CVT_F, and
! R_EFF_CVT_R. Transmission controller mode can be specified with the open-loop
! function MODE_TRANS and transmission gear can be specified with the open-loop
! function GEAR_TRANS.

INSTALL_TRANSMISSION  ! VS Command to install the supporting features for a
                      ! transmission in the powertrain
OPT_TRANS_INTERNAL  1 ! Transmission model: 1 -> internal, 0 -> external [I]
OPT_TR_GEAR_INTERNAL 1 ! Transmission gear ratio and inertia: 1 -> up to 7 gears, 2
                       ! -> continuously variable (CVT), 0 -> external model [I]
NGEARS              7 ! Number of gears in transmission [I]
OPT_SHIFT_INTERNAL  1 ! Gear shift command model: 1 -> internal, 0 -> external [I]
ITC_OUTPUT_SHAFT 0.002 ; kg-m2 ! Spin inertia of torque converter output shaft [I]
ITR_NEUTRAL      0.02 ; kg-m2 ! Neutral-gear spin inertia of transmission [I]
ITR_REVERSE      0.02 ; kg-m2 ! Reverse-gear spin inertia of transmission [I]
R_EFF_TR_F_REVERSE 0.99 ; - ! Reverse-gear forward efficiency of transmission [I]
R_EFF_TR_R_REVERSE 0.99 ; - ! Reverse-gear reverse efficiency of transmission [I]
R_GEAR_TR_REVERSE -2.6 ; - ! Reverse-gear ratio of transmission (should be negative)
                       ! [I]
T_SHIFT          0.05 ; s ! Shift duration in closed-loop transmission control [I]
ITR(1)           0.02 ; kg-m2 ! 1st-gear transmission spin inertia at output shaft
                      ! (Transmission spin inertias do not include the torque
                      ! converter) [I]
R_EFF_TR_F(1)    0.99 ; - ! 1st-gear forward efficiency of transmission [I]
R_EFF_TR_R(1)    0.99 ; - ! 1st-gear reverse efficiency of transmission [I]
R_GEAR_TR(1)      2.6 ; - ! 1st-gear ratio of transmission [I]
ITR(2)           0.02 ; kg-m2 ! 2nd-gear trans. spin inertia at output shaft
R_EFF_TR_F(2)    0.99 ; - ! 2nd-gear forward efficiency of transmission
R_EFF_TR_R(2)    0.99 ; - ! 2nd-gear reverse efficiency of transmission
R_GEAR_TR(2)      1.9 ; - ! 2nd-gear ratio of transmission
ITR(3)           0.02 ; kg-m2 ! 3rd-gear trans. spin inertia at output shaft
R_EFF_TR_F(3)    0.99 ; - ! 3rd-gear forward efficiency of transmission
R_EFF_TR_R(3)    0.99 ; - ! 3rd-gear reverse efficiency of transmission
R_GEAR_TR(3)      1.6 ; - ! 3rd-gear ratio of transmission
ITR(4)           0.02 ; kg-m2 ! 4th-gear trans. spin inertia at output shaft
R_EFF_TR_F(4)    0.99 ; - ! 4th-gear forward efficiency of transmission
R_EFF_TR_R(4)    0.99 ; - ! 4th-gear reverse efficiency of transmission
R_GEAR_TR(4)      1.4 ; - ! 4th-gear ratio of transmission
ITR(5)           0.02 ; kg-m2 ! 5th-gear trans. spin inertia at output shaft
R_EFF_TR_F(5)    0.99 ; - ! 5th-gear forward efficiency of transmission
R_EFF_TR_R(5)    0.99 ; - ! 5th-gear reverse efficiency of transmission
R_GEAR_TR(5)      1.2 ; - ! 5th-gear ratio of transmission
ITR(6)           0.02 ; kg-m2 ! 6th-gear trans. spin inertia at output shaft
R_EFF_TR_F(6)    0.99 ; - ! 6th-gear forward efficiency of transmission
R_EFF_TR_R(6)    0.99 ; - ! 6th-gear reverse efficiency of transmission
R_GEAR_TR(6)      1.1 ; - ! 6th-gear ratio of transmission
ITR(7)           0.02 ; kg-m2 ! 7th-gear trans. spin inertia at output shaft
R_EFF_TR_F(7)    0.99 ; - ! 7th-gear forward efficiency of transmission
R_EFF_TR_R(7)    0.99 ; - ! 7th-gear reverse efficiency of transmission
R_GEAR_TR(7)      0.8 ; - ! 7th-gear ratio of transmission
```

Figure 22. Parameters for transmission listed in the Echo file.

Figure 23 shows the two states of the transmission. The top diagram represents the geared position (either reverse or any forward gear shift position and the bottom diagram represents the neutral position. When the transmission is geared, it has one degree-of-freedom, in which the torque transfer devise output shaft to gearbox output shaft is treated as one body having one rotational motion. When the transmission is in neutral, it has an additional degree-of-freedom — the spin of the torque transfer device output shaft.

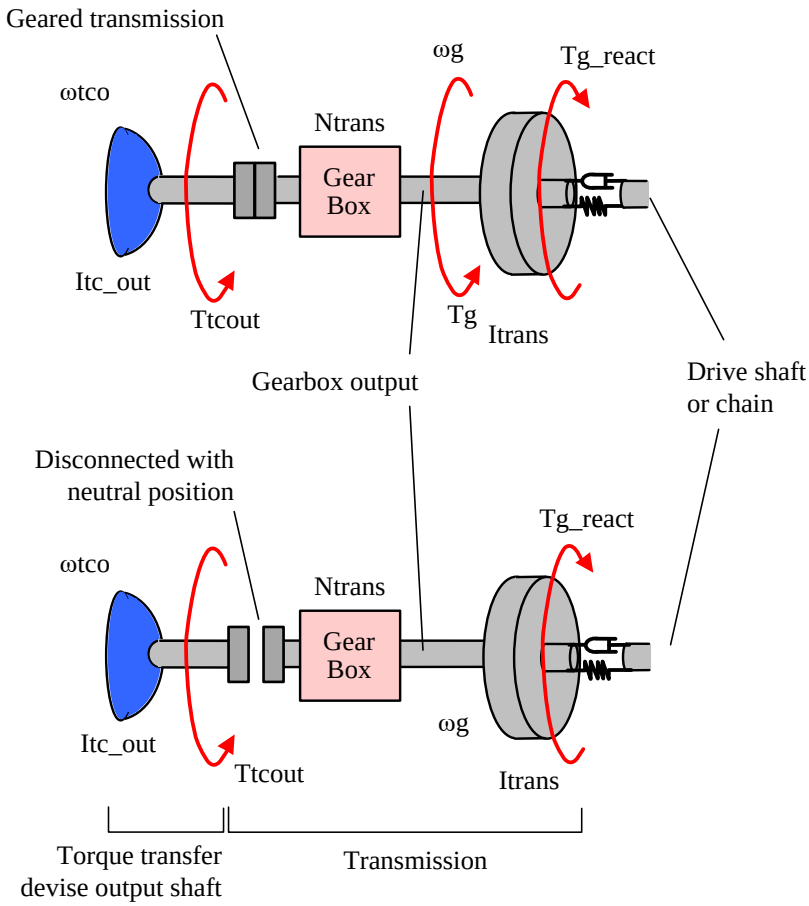


Figure 23. The BikeSim transmission model: geared transmission (top diagram) and disconnected with neutral position (bottom diagram).

The equations for the spin and torque regarding the transmission vary with whether geared or neutral, or whether the main clutch is engaged or disengaged (total 4 cases) as described below.

Cases of geared transmission

When the transmission is geared, the transmission gear ratio (T_{rams}) is associated with either the selected gear number or continuously variable. The output torque of the transmission gearbox (T_g) is given by the torque transfer device output torque (T_{tcout}), gear ratio and efficiency depending on whether the engine drives the wheel or the wheels drive the engine, such as:

$$\begin{aligned} T_g &= N_{trans} \times T_{tcout} \times E_{trans} \quad (T_{tcout}{}^{*} \geq 0) \quad or \\ T_g &= N_{trans} \times T_{tcout} \,/\, E_{trans_rev} \quad (T_{tcout}{}^{*} < 0) \end{aligned} \tag{18}$$

where E_{trans} is the efficiency (multiplying) when the engine drives the wheels. On the other hand, E_{trans_rev} is the efficiency (dividing) when the wheels drive the engine. Those two efficiencies are switched depending on the direction of the torque through the transmission.

Note All of the efficiency parameters used in the following descriptions have two cases, switched by the direction of T_{tcout}^{*}, which is compensated by engine inertial term when the clutch is locked in equation 15.

The gearbox output speed is given by the integration of the following differential equations:

If clutch is slipping

$$\dot{\omega}_g = \alpha_g = \frac{T_g - T_{g_react}}{I_{tc_out} \cdot N_{trans}^{\ 2} + I_{trans} + I_{trans_drv}}$$

else (clutch is locked) (19)

$$\dot{\omega}_g = \alpha_g = \frac{T_g - T_{g_react}}{\left\{(I_e \cdot N_{primary}^{\ 2}) + I_{tc_in} + I_{tc_out}\right\} \cdot N_{trans}^{\ 2} + I_{trans} + I_{trans_drv}}$$

where in case of the locked clutch, the engine, torque transfer devise output shaft and gearbox are all treated as one unit ($T_{tcout} = T_{tcout}^{*}$).

Note I_{trans_drv} is the inertia of a part of drive system which is either the gearbox sprocket (for chain drive) or a half of the driveshaft (for shaft drive). T_{g_react} is the reaction torque through the drive system. See more details in the sections **Shaft Drive Model** and **Chain Drive Model .**

The angular speed of the torque transfer devise output, ω_{tco}, is calculated by the transmission output angular speed, ω_g, multiplied by the transmission gear ratio, such as:

$$\omega_{tco} = N_{trans} \times \omega_g \tag{20}$$

Cases of neutral transmission

On the other hand, if the transmission is in neutral position (see the bottom diagram of Figure 23), the gearbox output speed is:

$$\dot{\omega}_g = \alpha_g = \frac{-T_{g_react}}{I_{trans} + I_{trans_drv}} \tag{21}$$

Furthermore, the output shaft of the torque transfer device has an independent degree-of-freedom. Its speed is given by the following equations.

$$\begin{aligned}
&\text{If clutch is slipping} \\
&\dot{\omega}_{tco} = \alpha_{tco} = \frac{T_{tcout}}{I_{tc_out}} \\
&\text{else (clutch is locked)} \\
&\dot{\omega}_{tco} = \alpha_{tco} = \frac{T_e \cdot N_{primary}}{I_e \cdot N_{primary}{}^2 + I_{tc_in} + I_{tc_out}}
\end{aligned} \tag{22}$$

For any case (whether transmission is geared or neutral), the gearbox input and output rotation angles are given by the differential equations, such as:

$$\begin{aligned}
\dot{\phi}_{tco} &= \omega_{tco} \\
\dot{\phi}_g &= \omega_g
\end{aligned} \tag{23}$$

Shift schedule

The gear of the transmission is determined in part by the transmission mode of operation. The mode is an open-loop function of time specified on the screen **Control: Automatic Gear shifting**. When the mode is -1, the reverse gear of the transmission is invoked. When the mode is 0, the transmission is in neutral: its output torque is zero. A mode of 1 means that the transmission gear is determined as an open-loop function of time as specified on the screen **Control: Manual Gear shifting**. Modes 2 through 7 cause the gears to be changed automatically. The details of the mode and gear settings are described in the separate documentation, **Rider Controls (Motorcycle)**.

The BikeSim automatic gear shifting mode uses look-up tables to determine when to shift up or down.

The screen shown in Figure 24 is used to specify the upshift boundary by table look-up of transmission speed as a function of throttle position. A nearly identical screen is used for the downshift boundary. Also, some special types of bike have a hydraulic torque converter, which may have a lockup clutch. In this case, locking and unlocking control schedules are specified by similar screens. Table 4 summarizes those schedule table screens.

Upshifting occurs when the transmission speed rises above a specified level or a throttle position drops below a specified level. Figure 25 shows an example upshift diagram with an upshift boundary from 3rd to 4th gear. In this figure, the current throttle position is 0.6 (60%) and transmission output speed is 2000 rpm with the current gear position of 3rd gear. The upshift will happen if the throttle position is reduced below around 0.38 keeping the same level of transmission speed, or the transmission speed increases more than 2600 rpm keeping the same level of throttle position.

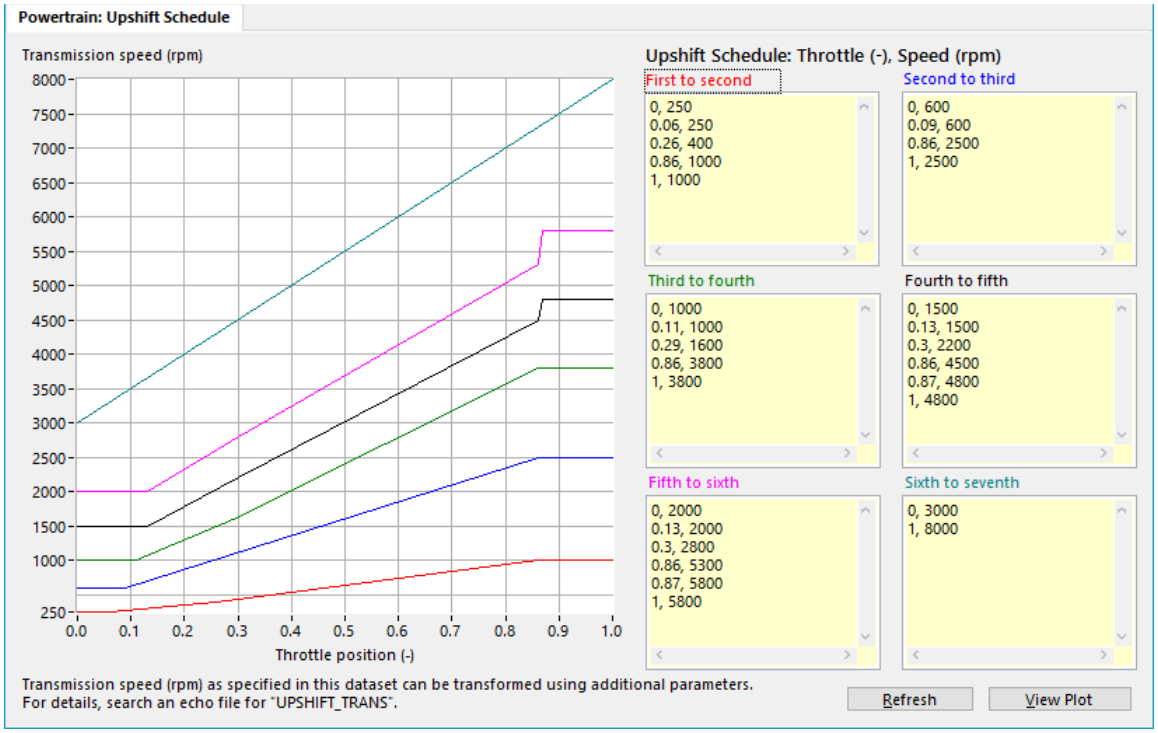


Figure 24. Shift schedule screen.

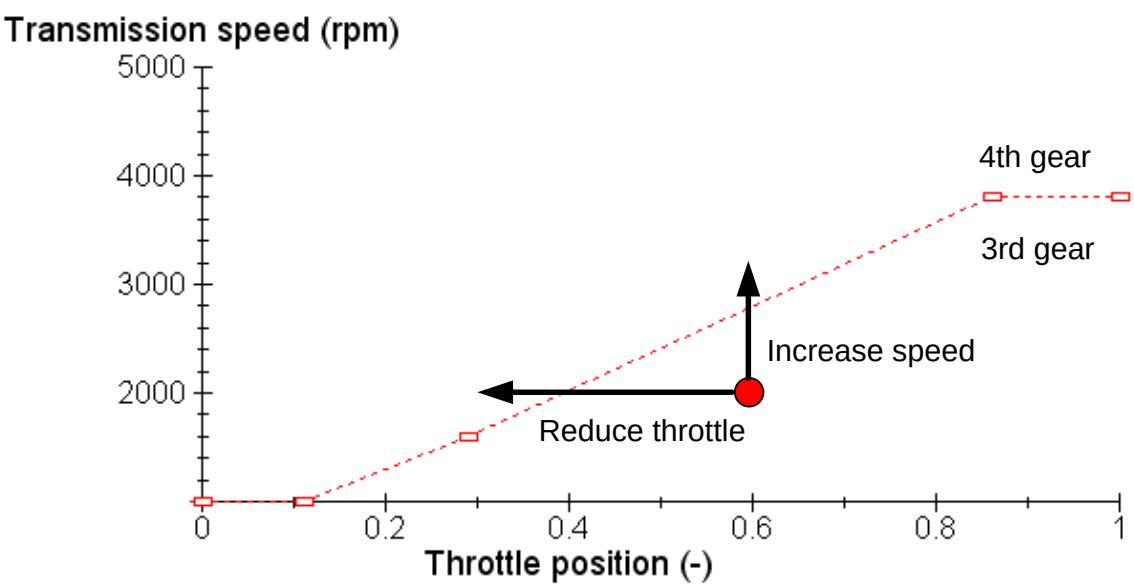


Figure 25. Example upshift diagram.

Downshifting occurs when the transmission speed decrease below a specified level or a throttle position rises above a specified level. Figure 26 shows an example downshift diagram with an downshift boundary from 4th to 3rd gear. In this figure, the current throttle position is 0.6 (60%) and transmission output speed is 2000 rpm. In this example, a downshift will occur if the throttle

position is increased above 0.82 keeping the same level of transmission speed, or if the transmission speed drops less than 1200 rpm keeping the same level of throttle position.

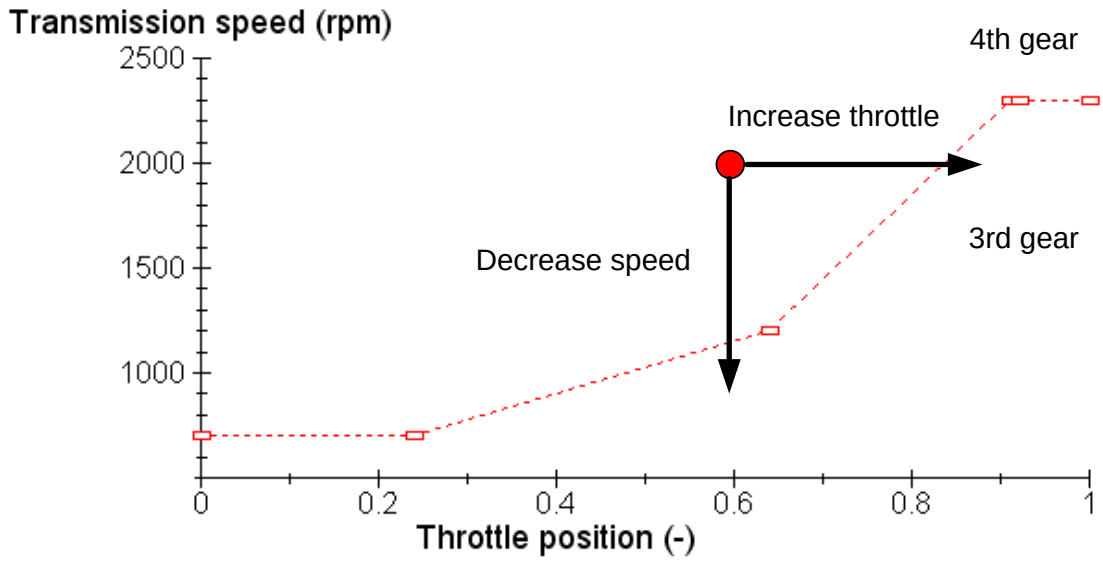


Figure 26. Example downshift diagram.

Transmission (7 Gears) Screen

Figure 27 shows the **Transmission (7 Gears)** screen that assembles parameters and links to shifting Configurable Functions.

① Drop-down list for selecting an external or internal shift schedule (keyword = `OPT_SHIFT_INTERNAL`). The data link for the upshift schedule② and downshift schedule ③ are hidden when the external shift schedule is selected.

② Link to **Powertrain: Upshift Schedule** datasets. The root keyword for the associated table is `UPSHIFT_TRANS` (Table 4.)

③ Link to **Powertrain: Downshift Schedule** datasets. The root keyword for the associated table is `DOWNSHIFT_TRANS` (Table 4.)

④ Checkbox to include a lockup clutch in the torque converter of an automatic transmission. Checking the box reveals two links (⑤ and ⑥) to define locking and unlocking control parameters for the lockup clutch (Figure 28). Also, the lockup clutch parameters (⑦ and ⑧) are used. This box must be unchecked when a manual clutch is linked on the main **Powertrain** screen.

⑤ Link to a **Powertrain: Lockup Schedule** datasets. The root keyword for the associated table is `LOCK_AT` (Table 4.)

⑥ Link to a **Powertrain: Unlock Schedule** datasets. The root keyword for the associated table is `UNLOCK_AT` (Table 4.)

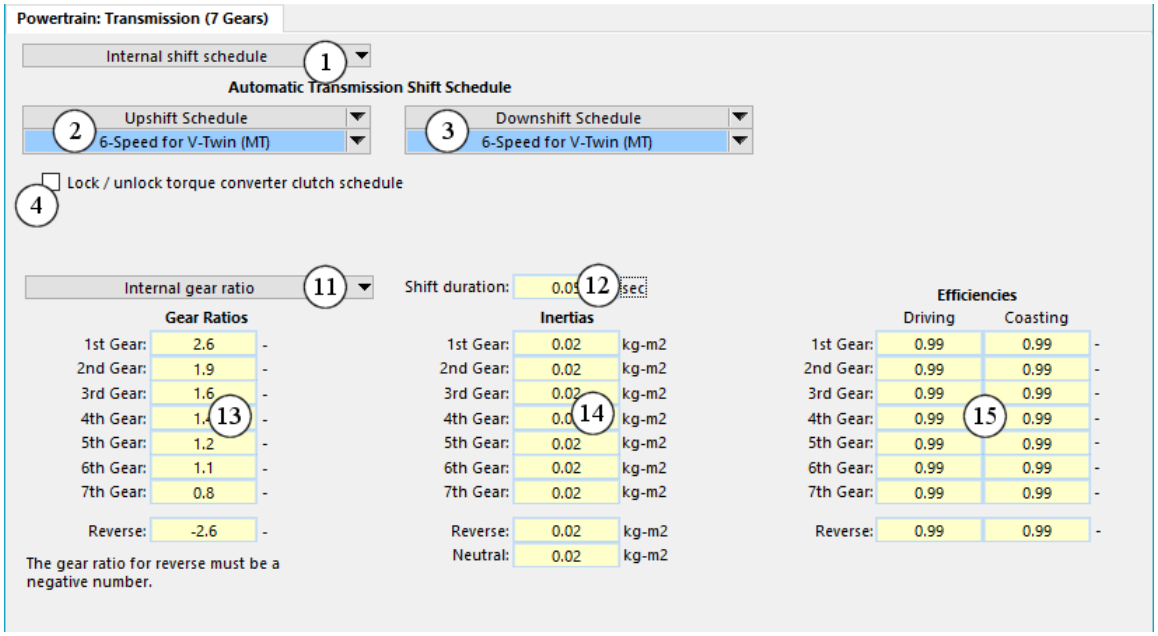


Figure 27. Screen for a transmission.

Table 4. Summary of gear shift and lock-up clutch control keywords.

Library Screen Name	Root Keyword	Index	Description
Powertrain: Upshift Schedule	`UPSHIFT_TRANS`	`IGEAR`	Gear Upshift schedule using throttle position and transmission output speed.
Powertrain: Downshift Schedule	`DOWNSHIFT_TRANS`	`IGEAR`	Gear downshift schedule using throttle position and transmission output speed.
Powertrain: Lockup Schedule	`LOCK_AT`	`IGEAR`	Torque converter lockup schedule using throttle position and transmission output speed.
Powertrain: Unlock Schedule	`UNLOCK_AT`	`IGEAR`	Torque converter unlock schedule using throttle position and transmission output speed.

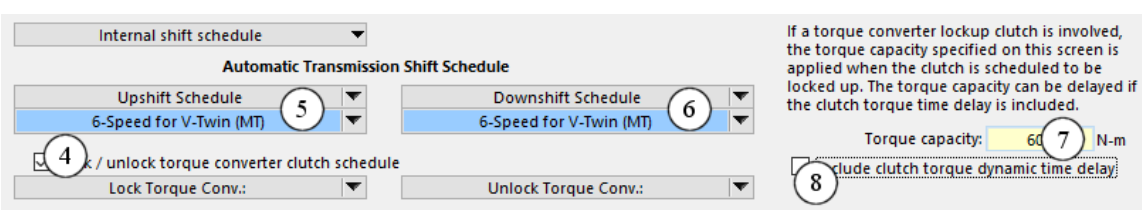
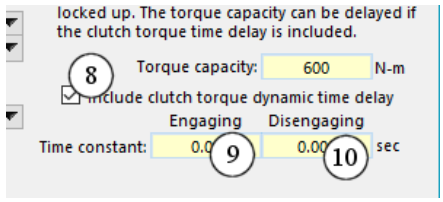


Figure 28. Additional controls for the lock/unlock clutch schedule.

⑦ Maximum torque capacity of the lockup clutch (keywords = `M_LOCKUP_CLUTCH_CAP`).

⑧ Checkbox for using a 1st order time delay of the clutch torque capacity (keyword = `OPT_CLUTCH_DELAY`). When checked, extra data fields (⑨ and ⑩) are shown that specify the time constants. If not checked, the clutch torque responds instantly without delay.



⑨ Dynamic time constant for engaging the clutch (keyword = `TC_CLUTCH_ENGAGE`).

⑩ Dynamic time constant for disengaging the clutch (keyword = `TC_CLUTCH_DISENGAGE`).

⑪ Drop-down list for selecting external or internal transmission gear ratios (keyword = `OPT_TR_GEAR_INTERNAL`).

⑫ Shift duration, which is the transition time to change from one gear to another (keyword = `T_SHIFT`)

⑬ Driveline gear ratios (keywords = `R_GEAR_TR_REVERSE`, `R_GEAR_TR(IGEAR)`). The forward gear ratios must be positive and the reverse gear ratio must be negative.

⑭ Driveline inertias (keywords = `ITR_REVERSE`, `ITR_NEUTRAL`, `ITR(IGEAR)`). All values should be positive. For front and rear-wheel drive, the equivalent inertias are the inertial loads placed on the respective differential. For both part-time and full-time four-wheel drive, the equivalent inertias are the inertial loads placed on the transfer case, and the drive shafts are assumed to have zero inertia.

⑮ Natural (forward) and inverse (reverse) efficiencies for each gear (keywords = `R_EFF_TR_F(IGEAR)`, `R_EFF_TR_F_REVERSE`, `R_EFF_TR_R(IGEAR)`, `R_EFF_TR_R_REVERSE`).

Continuously Variable Transmission (CVT)

When the transmission is continuously variable (CVT), the primary gear ratio ($N_{primary}$) is continuously variable as a function of the throttle position and the CVT output speed (= clutch input speed), such as:

$$N_{primary_inst} = f_{cvt_gear}(throttle, \omega_{tcin}) \tag{24}$$

where $N_{primary_inst}$ is the instant gear ratio which does not involve any time delay. The actual gear ratio applied to the primary is expressed by a first-order dynamic time delay such as:

$$\dot{N}_{primary} = \frac{N_{primary_inst} - N_{primary}}{T_{c_cvt}} \tag{25}$$

where T_{c_cvt} is the dynamic time constant, and $N_{primary}$ is integrated and updated for every numerical calculation time step. In turn, the CVT efficiencies (E_{cvt} and E_{cvt_rev}) and CVT inertia (I_{cvt}) are also continuously variable as the functions of the current primary gear ratio, such as:

$$\begin{aligned} &E_{cvt} = f_{cvt_efficiency}(N_{primary}), \\ &E_{cvt_rev} = f_{cvt_efficiency_rev}(N_{primary}), \text{ and} \\ &I_{cvt} = I_{cvt_input_pulley} \times N_{primary}^{2} + I_{cvt_output_pulley}, \end{aligned} \tag{26}$$

where $I_{cvt_input_pulley}$ and $I_{cvt_output_pulley}$ are the inertial of the input and output pulley of the CVT, respectively. CVT inertia (I_{cvt}) is added on the clutch input shaft inertia (I_{tc_in}). CVT efficiencies are always applied to the engine output torque through the primary gear.

Because the gear ratio is differentiable with time, the angular acceleration of the engine can be:

$$\dot{\omega}_e = N_{primary}\ \dot{\omega}_{tcin} + \dot{N}_{primary}\ \omega_{tcin} \tag{27}$$

which yields the engine speed depending on the clutch state, such as:

$$\begin{aligned} &\text{If clutch is slipping} \\ &\dot{\omega}_e = \alpha_e = \frac{T_{e_in} - T_{e_out} - (I_e + I_{cvt_input_pulley}) \cdot \dot{N}_{primary} \cdot \omega_{tcin}}{I_e + \dfrac{I_{tc_in} + I_{cvt}}{N_{primary}^{2}}} \\ &\text{else (clutch is locked)} \\ &\dot{\omega}_e = \alpha_e = N_{primary}\ \dot{\omega}_{tco} + \dot{N}_{primary}\ \omega_{tco} \\ &T_{tcin} = T_{tcout} = T_e \cdot N_{primary} - (I_e + I_{cvt_input_pulley}) \cdot N_{primary} \cdot \dot{N}_{primary} \cdot \omega_{tco} \end{aligned} \tag{28}$$

Equation 28 is like equations 3 and 14 but involves an additional term with the time derivative of primary gear ratio. If the primary gear ratio continuously varies (time differentiable) such as CVT, the engine speed is calculated by equation 28 instead of equations 3 and 14.

CVT parameters can be set on either the **Powertrain: Shaft Drive** or **Powertrain: Chain Drive** screen. The CVT gear ratio and efficiencies can be also linked from those screens. Table 5 shows the summary of the data tables.

Table 5. Summary of the continuously variable gear ratio and efficiencies control keywords.

Library Screen Name	**Root Keyword**	**Index**	**Description**
Powertrain: CVT Gear Ratio	`R_GEAR_CVT`	`ICVT`	CVT gear ratio as a function of throttle position and transmission output speed. The index refers to either the drive (forward) or reverse.
Powertrain: CVT Efficiencies	`R_EFF_CVT_F` and `R_EFF_CVT_R`	`ICVT`	CVT driving and coasting efficiencies, respectively, as functions of the transmission gear ratio. The index refers to either the drive (forward) or reverse.

Shaft Drive Model

Driveshaft and final Gear

Angular speed of the driveshaft (wheel side) ω_{shaft} is defined by the angular speed of the rear wheel and the final gear ratio, N_{final}, such as:

$$\omega_{shaft} = \omega_{wh_R} \times N_{final} \tag{29}$$

Rotation angle of the driveshaft (wheel side) ϕ_{shaft} is defined by the rotational angle of the rear wheel and the final gear ratio N_{final}, such as:

$$\phi_{shaft} = \phi_{wh_R} \times N_{final} \tag{30}$$

Driveshaft torque is calculated using torsional stiffness K_{shaft} and damping D_{shaft} of the driveshaft. However, when the torque is negative (the wheel drives the engine), the torsion involves a backlash, such as:

$$\begin{aligned} T_{shaft} &= K_{shaft} \times (\phi_g - \phi_{shaft}) + D_{shaft} \times (\omega_g - \omega_{shaft}) \qquad \left(T_{shaft} \geq 0\right) \\ T_{shaft} &= K_{shaft} \times (\phi_g - \phi_{shaft} + \phi_{backlash}) + D_{shaft} \times (\omega_g - \omega_{shaft}) \qquad \left(T_{shaft} < 0\right) \end{aligned} \tag{31}$$

The driveshaft torque is used to calculate the transmission output speed as a reaction torque in equation 19. Also, half of the driveshaft inertia is assigned to I_{trans_drv}, which is added on the transmission inertia in equations 12 and 19, such as:

$$\begin{aligned} T_{g_react} &= T_{shaft} \\ I_{trans_drv} &= I_{shaft} / 2 \end{aligned} \tag{32}$$

Driving Wheel Calculation

Driveshaft torque is transferred to the rear wheel through the final gear:

$$M_{drv_R} = T_{shaft} \times N_{final} \times E_{final} \tag{33}$$

where E_{final} is the efficiency of the final gear. Different values for the parameter E_{final} are used depending on whether the engine drives the wheel or the wheel drives the engine.

The angular acceleration of the rear wheel is:

$$\dot{\omega}_{wh_R} = \frac{M_{drv_R} + M_{y_resis} - \left\{F_{x_R}(R_{Eff} - \Delta_T) + M_{bk_R}\right\}}{I_{wR} + (I_{shaft} \times N_{final}{}^2 / 2)} \tag{34}$$

where F_{x_R} is the longitudinal force of rear tire, M_{bk_R} is the braking torque of a rear wheel, M_{y_resis} is the rolling resistance torque, I_{wR} is the moment of inertia of a rear wheel, R_{Eff} is the effective rolling radius and Δ_T is the tire deflection of rear tire.

User Settings

These driveshaft parameters are defined on the earlier section *Assembling Powertrain Components*, Figure 4, page 3).

Chain Drive Model

Chain Mechanism

The chain functions as a tension link between the points p_{gu} on the gearbox sprocket and p_{wu} on the rear wheel sprocket when the engine is driving the wheel. When the wheel is driving the engine, the chain functions as a tension link between p_{wl} and p_{gl} (Figure 29). Some slack or clearance is allowed between driving and overrunning, and the chain is treated as elastic and damped [1,2]. During the simulation, the direction of the vector $\mathbf{r}_{sw}$ always changes according to the wheel stroke. Therefore, the points (p_{gu}, p_{wu}, p_{gl} and p_{wl}) are defined as moving points on the sprocket bodies. The locations of those four points are defined by the angles, η_u and η_l, which are given by the following geometric analysis.

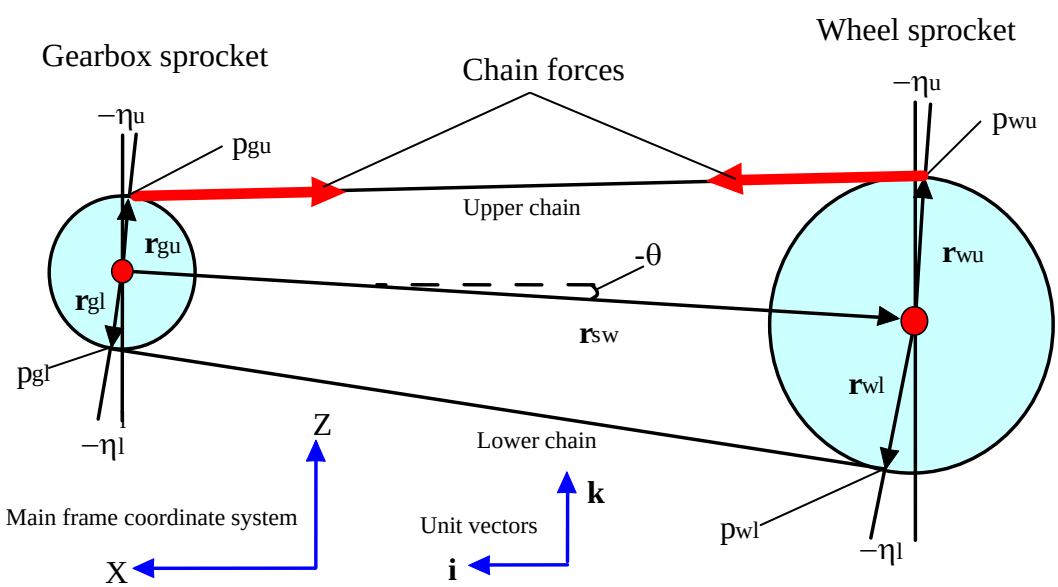


Figure 29. Geometry of the chain drive system.

Note	The original analysis was presented by Robin Sharp, Simos Evangelou and David Limebeer in their literature [1,2]. However, the equations are simplified here to match more generic suspension kinematics in BikeSim.

In Figure 29, these relationships are apparent:

$$\begin{aligned}
\mathbf{r}_{sw} &= -r_{sw}\cos\theta\,\mathbf{i} + r_{sw}\sin\theta\,\mathbf{k} \\
\mathbf{r}_{wu} &= r_w \sin\eta_u\,\mathbf{i} + r_w\cos\eta_u\,\mathbf{k} \\
\mathbf{r}_{gu} &= r_g \sin\eta_u\,\mathbf{i} + r_g\cos\eta_u\,\mathbf{k} \\
\mathbf{r}_{wl} &= -r_w \sin\eta_l\,\mathbf{i} - r_w\cos\eta_l\,\mathbf{k} \\
\mathbf{r}_{gl} &= -r_g \sin\eta_l\,\mathbf{i} - r_g\cos\eta_l\,\mathbf{k}
\end{aligned} \tag{35}$$

The chain lines are perpendicular to the radius vector:

$$\begin{aligned}
(-\mathbf{r}_{wu} - \mathbf{r}_{sw} + \mathbf{r}_{gu})\cdot\mathbf{r}_{gu} &= 0 \\
(-\mathbf{r}_{wl} - \mathbf{r}_{sw} + \mathbf{r}_{gl})\cdot\mathbf{r}_{gl} &= 0
\end{aligned} \tag{36}$$

Substitution of Equations 35 into 36 yields:

$$\eta_u = \sin^{-1}\left(\frac{r_w - r_g}{r_{sw}}\right) + \theta \tag{37}$$

$$\eta_l = \sin^{-1}\left(\frac{r_g - r_w}{r_{sw}}\right) + \theta \tag{38}$$

Chain Force Calculation

The upper chain run is in tension if $(r_g\phi_g - r_w\phi_{wh_R} + \delta l_u)$ is positive, with the first two terms accounting for the spinning of the sprockets and the third for the increase in separation of p_{gu} and p_{wu} due to suspension motion. Similarly, the lower chain run is in tension if $(-r_g\phi_g + r_w\phi_{wh_R} + \delta l_u - s_l)$ is positive, in which S_l represents the chain slack. If the upper or lower chain is in tension, it applies a force between the end points using a spring and damper.

Gearbox Sprocket Calculation

The gearbox sprocket is directly connected to the transmission output shaft. Therefore, the sprocket speed is calculated by the differential equations 19, in which the reaction to the transmission through the upper chain, when the engine drives the wheel, is given by:

$$T_{g_react} = F_{upper\ chain} \times r_g\,, \tag{39}$$

or through the lower chain, when the wheel drives the engine, is given by:

$$T_{g_react} = -F_{lower\ chain} \times r_g\,. \tag{40}$$

Also, the sprocket inertia (I_{sprkt}) is assigned to I_{trans_drv} which is added on the transmission inertia in equations 12 and 19, such as: $I_{trans_drv} = I_{sprkt}$.

Driving Wheel Calculation

Torque is transferred to the rear wheel through the chain force. The angular acceleration of the rear wheel is when the engine drives the wheel:

$$\dot{\omega}_{wh_R} = \frac{F_{upper\ chain} \times r_w + M_{y_resis} - \{F_{x_R}(R_{Eff} - \Delta_T) + M_{bk_R}\}}{I_{wR}}, \tag{41}$$

or when the wheel drives the engine:

$$\dot{\omega}_{wh_R} = \frac{-F_{lower\ chain} \times r_w + M_{y_resis} - \{F_{x_R}(R_{Eff} - \Delta_T) + M_{bk_R}\}}{I_{wR}}, \tag{42}$$

where F_{x_R} is the longitudinal force of rear tire, M_{bk_R} is the braking torque of a rear wheel, M_{y_resis} is the rolling resistance torque, I_{wR} is the moment of inertia of a rear wheel, R_{Eff} is the effective rolling radius and Δ_T is the tire deflection of rear tire.

User Settings

These driveshaft parameters are defined on the earlier section *Assembling Powertrain Components* and Figure 4 (page 3).

Import Variables

Import variables offer a method to control aspects of the powertrain model via equations or external software (e.g., supplying engine torque values externally). Typically, internal variables used by the model can be combined with user-defined values according to one of three user-specified modes: ADD, REPLACE, and MULTIPLY.

> **Note** For more information on Import variables, see the *VS Commands Reference Manual*.

Due to the modular nature of the powertrain model, not all import modes (ADD/REPLACE/MULTIPLY) are available in all model configurations. Some Import variables are specialized and only have an effect in specific contexts.

The following table lists all powertrain Import variables that can have different behavior than the standard ADD/REPLACE/MULTIPLY modes.

Table 6. Powertrain Import variables.

Import Variable	Description of Behavior
`IMP_ESC_ENGINE_CON_STATE`	This import keyword uses integer input only `[0, 1, 2]` to switch between modes. All import behavior is REPLACE.

Table 6. Powertrain Import variables.

Import Variable	**Description of Behavior**
`IMP_MENGINE_ESC_REQUEST`	This import keyword has no internal equivalent and is only used in tandem with `IMP_ESC_ENGINE_CON_STATE`. All import behavior is REPLACE.
`IMP_MENGINE`	This import keyword is affected by the modular powertrain. When `[OPT_ENGINE_INTERNAL=0]`, the import behavior is REPLACE. When `[OPT_ENGINE_INTERNAL=1]`, the import behaves normally for ADD, MULTIPLY, and REPLACE.
`IMP_AV_ENG`	This import keyword is affected by the modular powertrain. When `[OPT_ENGINE_INTERNAL=0]`, the import behavior is REPLACE. When `[OPT_ENGINE_INTERNAL=1]`, this import has no effect.
`IMP_R_GEAR_TR`	This import keyword is affected by the modular powertrain. When `[OPT_TRANS_INTERNAL=0]`, the import behavior is REPLACE. When `[OPT_TRANS_INTERNAL=1]`, the import behaves normally for ADD, MULTIPLY, and REPLACE.
`IMP_INERTIA_TR`	This import keyword is affected by the modular powertrain. When `[OPT_TRANS_INTERNAL=0]`, the import behavior is REPLACE. When `[OPT_TRANS_INTERNAL=1]`, the import behaves normally for ADD, MULTIPLY, and REPLACE.
`IMP_M_OUT_TR`	This import keyword is affected by the modular powertrain. When `[OPT_TRANS_INTERNAL=0]`, the import behavior is REPLACE. When `[OPT_TRANS_INTERNAL=1]`, the import behaves normally for ADD, MULTIPLY, and REPLACE.

Table 6. Powertrain Import variables.

Import Variable	Description of Behavior
IMP_MENG_REACT	This import keyword is affected by the modular powertrain. When [OPT_PWR_CPL_INTERNAL=0], the import behavior is REPLACE. When [OPT_PWR_CPL_INTERNAL=1 and OPT_CLUTCH=0], the import behavior is ADD. When [OPT_PWR_CPL_INTERNAL=1 and OPT_CLUTCH=1], the import behaves normally for ADD, MULTIPLY, and REPLACE. When [OPT_PWR_CPL_INTERNAL=1 and OPT_CLUTCH=2], the import behavior is ADD.
IMP_M_OUT_TC	This import keyword is affected by the modular powertrain. When [OPT_PWR_CPL_INTERNAL=0], the import behavior is REPLACE. When [OPT_PWR_CPL_INTERNAL=1], the import behavior is ADD.
IMP_MYDR_1	The import behavior is always REPLACE.
IMP_MYDR_2	This import keyword is affected by the modular powertrain. When [OPT_PT=1 and OPT_FINAL_DRIVE_INTERNAL=1], the import behaves normally for ADD, MULTIPLY, and REPLACE. Otherwise, the import behavior is REPLACE or has no effect.
IMP_MYDR_3	When [OPT_PT=0], the import has no effect. Otherwise, the import behavior is REPLACE.

References

1. Sharp, R. S., Evangelou, S. and Limebeer, D. J. N., “Advances in the modelling of motorcycle dynamics”, Multibody System Dynamics, 12(3), 2004, 251-283.

2. Sharp, R. S., Evangelou, S. and Limebeer, D. J. N., “Multibody aspects of motorcycle modelling with special reference to Autosim”, Advances in Computational Multibody Systems, J. G. Ambrósio (Ed.), Springer-Verlag, Dordrecht, The Netherlands, 2005, 45-68.

Appendix

Figure 30 shows the torques acting on both sides of a locked clutch.

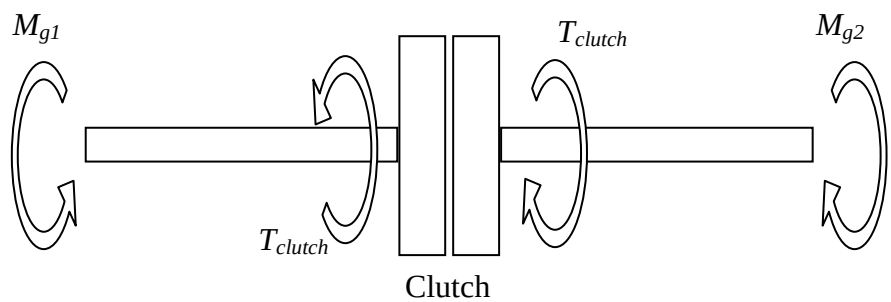


Figure 30. Torques acting on both sides of a locked clutch.

M_{g1} and M_{g2} are the torque acting on each side of the shaft. If this is the clutch between the engine and transmission, M_{g1} is the engine torque applied on the crankshaft while M_{g2} is the torque load from the transmission. On the other hand, if this is the clutch between left and right wheels (differential clutch), they are the torque load on the wheels due to the tire forces/moment and brake torques.

The angular speed on each side is described by:

$$\begin{aligned} M_{g1} - T_{clutch} &= I_{axle_1} \cdot \dot{\omega}_{g1} \\ M_{g2} + T_{clutch} &= I_{axle_2} \cdot \dot{\omega}_{g2} \end{aligned} \tag{43}$$

If the clutch is locked, the angular acceleration of both sides are synchronized such as:

$$\dot{\omega}_{g1} = \dot{\omega}_{g2} \tag{44}$$

From equations 43 and 44,

$$M_{g2} + T_{clutch} = I_{axle_2} \cdot \left(\frac{M_{g1} - T_{clutch}}{I_{axle_1}} \right) \tag{45}$$

From equation 45, the torque loaded on the clutch when it is locked is:

$$T_{clutch_lock} = \frac{M_{g1} - M_{g2} \cdot \dfrac{I_{axle_1}}{I_{axle_2}}}{1 + \dfrac{I_{axle_1}}{I_{axle_2}}} \tag{46}$$

Therefore, if the torque load derived from equation 46 is smaller than the torque capacity of the clutch, the clutch can be considered as locked. On the other hand, if the torque load exceeds the torque capacity, the clutch will be slipping.

Notation

D_{shaft} — Torsional damping coefficient of driveshaft

D_{lock_clutch} — Torsional damping coefficient of clutch

$d\omega_{clutch}$ — Differential of angular speeds between input and output plate of mechanical clutch

$d\omega_{clutch_old}$ — Differential of angular speeds between input and output plate of mechanical clutch (at 1 calculation time step before)

δl_u — Upper chain length change from a nominal length (positive when extend)

δl_l — Lower chain length change from a nominal length (positive when extend)

E_{final} — Efficiency of final gear (shaft drive model)

ESC_state — ESC engine torque control state

E_{trans} — Efficiency of transmission (adopted in the calculation)

E_{trans_rev} — Efficiency of transmission (wheel driving engine)

$F_{upper\ chain}$ — Upper chain tension force

$F_{lower\ chain}$ — Lower chain tension force

F_{x_R} — Longitudinal force of rear tire

f_{cvt_gear} — CVT gear ratio, function of a throttle position and transmission output speed

$f_{cvt_efficiency}$ — CVT efficiency (driving), function of a transmission gear ratio

$f_{cvt_efficiency_rev}$ — CVT efficiency (coasting), function of a transmission gear ratio

f_{clutch_cap} — Maximum clutch capacity torque, function of a clutch pedal position

f_{et} — Engine torque, function of a throttle position and crankshaft rotational speed

f_{tki} — Torque capacity, function of the speed ratio of torque converter

f_{tr} — Torque ratio, function of speed ratio of torque converter

f_{ush} — Shift up timing (transmission output rotational speed and throttle position)

G_{P_ESC} — P gain of ECS/TCS engine torque controller

G_{I_ESC} — I gain of ECS/TCS engine torque controller

$I_{cvt_input_pulley}$ — Moment of inertia of the CVT input pulley

$I_{cvt_output_pulley}$ — Moment of inertia of the CVT output pulley

I_{cl_in} — Total moment of inertia from Engine to the clutch input plate expressed at the clutch position

I_{cl_out} — Total moment of inertia from the clutch output plate to transmission gear box sprocket (or half of driveshaft) expressed at the clutch position

I_e — Moment of inertia of Engine (all rotating parts)

I_{e_comb} — Total moment of inertia from engine to the clutch input plate expressed at the engine position

I_{sprkt} — Moment of inertia of gearbox sprocket (chain drive model)

I_{tc_in} — Moment of inertia of half of the clutch or torque converter (engine side)

I_{tc_out} — Moment of inertia of half of the clutch or torque converter (transmission side)

I_{trans} — Moment of inertia of all the rotating parts with the transmission. The inertial is represented just after the gearbox.

I_{trans_drv} — Moment of inertia of a part of the drive system either chain sprocket or driveshaft. The inertial is added on after the gearbox.

I_{shaft} — Moment of inertia of rear drive shaft

I_{wR} — Moment of inertia of a rear wheel

K_{Lock_clutch} — Torsional stiffness of the clutch

K_{shaft} — Torsional stiffness of the driveshaft

M_{bk_R} — Braking torque of a rear wheel

M_{drv_R} — Drive torque of rear wheel

M_{y_resis} — Rolling resistance of rear wheel

N_{final} — Final gear ratio (shaft drive)

$N_{primary}$ — Primary gear ratio

$N_{primary_inst}$ — Instant gear ratio of the CVT (without dynamic time delay) at the primary gear location

N_{trans} — Gear ratio of the transmission

p_{gu} — Point which the line from this point to gearbox sprocket center and upper chain is perpendicular.

p_{gl} — Point which the line from this point to gearbox sprocket center and lower chain is perpendicular.

p_{wu} — Point which the line from this point to wheel sprocket center and upper chain is perpendicular.

p_{wl} — Point which the line from this point to wheel sprocket center and lower chain is perpendicular.

R_{Eff} — Effective rolling radius

r_g — Gearbox sprocket radius

$\boldsymbol{r}_{gu}$ — Vector from the center of gearbox sprocket to upper chain force point (p_{gu})

$\boldsymbol{r}_{gl}$ — Vector from the center of gearbox sprocket to lower chain force point (p_{gl})

r_{sw} — Distance between gearbox sprocket center to rear wheel center

$\boldsymbol{r}_{sw}$ — Vector from the center of gearbox sprocket to rear wheel center

r_w — Wheel sprocket radius

$\boldsymbol{r}_{wu}$ — Vector from the center of rear wheel to upper chain force point (p_{wu})

$\boldsymbol{r}_{wl}$ — Vector from the center of rear wheel to lower chain force point (p_{wl})

R_{cl_lock} — Shaping factor used for the smooth transition in the torque reversal of the main clutch

S_{tc} — Speed ration of the torque converter

S_l — Chain slack

T_{c_cvt} — Dynamic time constant of CVT

T_{clutch} — Clutch torque

T_{clutch_cap} — Maximum torque capacity of the clutch

T_{clutch_lock} — Theoretical clutch torque, which is supposed to be locked

T_{e_in} — Engine torque applied to crank shaft from the piston cylinders

T_{e_out} — Reaction torque from clutch to engine

T_g — Transmission output torque

$T_{e_control}$ — Engine torque modulated by ESC/TCS

$T_{e_DRV_rq}$ — Driver requesting engine torque

$T_{e_ESC_rq}$ — Engine torque requested by ESC/TCS

T_{e_target} — ESC targeting engine torque

T_{g_react} — Transmission output torque reaction through the drive system

T_{tcin} — Input torque of the clutch

T_{tcout} — Output torque of the clutch

T_{shaft} — Driveshaft output torque

throttle — Throttle position

$V_{low_cl_lock}$ — Speed threshold for lock/slip condition of the main clutch

α_e — Engine angular acceleration

Δ_T — Tire deformation of rear tire

$\phi_{backlash}$ — Shaft-drive gear backlash angle

ϕ_e — Engine rotation angle

ϕ_g — Transmission output shaft rotation angle

ϕ_{shaft} — Driveshaft rear end angle

ϕ_{tcin} — Clutch input angle

ϕ_{tco} — Clutch output angle

ϕ_{wh_R} — Rear wheel rotation angle

θ — Angle of the vector (gearbox sprocket center to wheel center) to the horizontal of the main body coordinate system

η_u — Upper chain angle in the main body coordinate system

η_l — Lower chain angle in the main body coordinate system

ω_e — Engine angular speed

ω_g — Transmission output angular speed

ω_{shaft} — Driveshaft rear end angular speed

ω_{tcin} — Clutch input angular speed

ω_{tco} — Clutch output angular speed

ω_{wh_R} — Rear wheel angular speed